Directed Recovery Structures in Tape Computations

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Abstract

Depth reduction turns an oblivious tape computation of matrix transposition into an index code whose side information consists of bounded dependency cones. A coding lower bound is useful only if its hypotheses follow from those actual cones. For fixed-depth cones of linear-time native schedules, we prove the following structural statement: after conditioning on heavily reused input coordinates, the input–demand incidence graph has bounded degree, and only $O(L/\log d)$ matched input and output ports lie at any fixed graph distance, where $L = d^2 b$ is the bit length of a $d \times d$ matrix. The statement concerns the full syntactic dependency graph and requires no correctness assumption on the computation.

For bounded-degree directed incidence systems with this separation property, we prove a half-rate bound for linear storage codes over the complex numbers with fixed block size and uniformly bounded recovery coefficients. For arbitrary finite alphabets, we prove a half-rate bound under uniform hyperfiniteness. Native examples show that hyperfiniteness and separation of matched ports into distinct connected components cannot be imposed universally: inexpensive tape executions produce robust expanding incidence and many connected demands. Nevertheless, their full cone relations admit nonlinear coding bounds, including a positive rate deficit independent of the fixed depth parameter. An appendix gives matrix certificates over arbitrary fields and applies them to a separate expanding family in characteristic two.

These results identify a directed coding problem naturally forced by tape geometry. The general finite-alphabet coding bound, unrestricted cheap depth reduction, and an all-input treatment of adaptive schedules remain open; no unrestricted transposition time bound follows here.

The conditioning, fibre, projection, rectangle, threshold-pullback, port-color, component, separator, damaged-expander, rank-identity, matrix-label, and hyperfinite half-rate results are formalized in Lean. The paired-distance bound, self-demands, degree trimming, the depth-independent code deficit, and the expanding-family linear-code corollary are partially formalized. The remaining native-computation and complex-analytic arguments have written proofs and are not yet formalized.

Contributions. Hans Brende chose the problem and directed the research. Codex, a GPT-6-based assistant developed by OpenAI, developed the mathematical arguments, Lean formalizations, and text under his direction. Formal verification is recorded result by result; the remaining proofs are explicitly identified.

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1 Introduction

A tape machine can compute a local linear transformation cheaply without making the corresponding recovery graph simple. Conversely, a complicated recovery graph need not support a high-rate code. These two observations create an applicability problem for coding approaches to matrix transposition: one needs a structural condition that follows from the actual execution, together with a coding theorem that uses that condition without silently strengthening it.

This paper studies that interface. Its central object is a directed recovery relation, represented by a bipartite graph with two copies of every coordinate. The left copy is an input port; the right copy is the receiver demanding that coordinate. An edge records available side information. The condition supplied by tape geometry is that almost all matched ports are far apart. Ordinary bipartiteness of the incidence graph is automatic. Symmetry of the recovery relation, local bipartiteness after symmetrization, cheap separators, and disconnected matched ports are additional assertions, and some fail for inexpensive native executions.

1.1 The transposition problem and the coding interface

Fix a deterministic multitape Turing machine with a finite state set, a fixed finite work alphabet, a read-only two-way input tape, finitely many two-way work tapes, and an append-only binary output. The input consists of a $d \times d$ matrix written in row-major order, with

$$b(d) = \max\{1, \lceil \log_2 d \rceil\}, \qquad L = d^2 b(d).$$

The machine must output the transposed matrix in the same format. Statement (T) asks whether one such machine can perform this task in $O(L)$ time for all d. The earlier papers develop its relation to integer multiplication, causal coding, restricted machine architectures, and bit-versus-word transposition [5, 7, 6, 4]. The native incidence-distance theorem allows every entry width $b \geq 1$; the explicit machines below use the canonical width.

An oblivious machine has one head schedule and one set of output times at each input length. A *genie* reveals the whole values of selected computation steps. If deleting dependencies into those steps leaves bounded depth, each output is a function of the genie and a bounded set of raw input coordinates. If the machine transposes correctly, fixing the genie leaves a storage code for these side-information sets. A code-size deficit larger than the genie cost gives a contradiction. This is the elementary decomposition of [4, Theorem D]; we recall the required finite counting argument in Section 2.

The unresolved issue is not the counting argument. It is the structure and capacity of the full cone relations. A lower bound for linear codes alone does not control an arbitrary genie, and a graph theorem does not apply until its graph hypothesis has been justified from the execution.

1.2 Results and their scope

The main results follow one progression.

(1) **A structural theorem for actual cones.** For every fixed depth and degree bound, a linear-time native schedule has only $O(L/\log d)$ coordinates whose matched ports are within any fixed incidence distance. Conditioning on heavily used inputs makes both incidence degrees bounded at an arbitrarily small fixed coordinate cost. The proof realizes short incidence paths as bounded-congestion geometric transport along alternating forward and reverse copies of the *same* head schedule. Reversing geometric paths does not invert the computation's local functions.

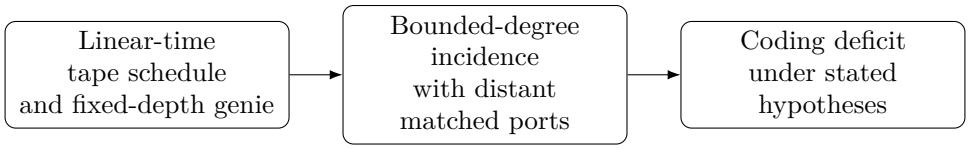


Figure 1: The interface developed here. The first arrow holds for actual fixed-depth cones after paid degree trimming. The second is proved in several specified regimes, not for all native finite-alphabet recovery systems.

(2) **A directed analytic coding theorem.** Bounded-degree incidence systems satisfying this paired-distance condition have complex linear storage rate at most $1/2 + o(1)$ when block size and recovery coefficients are uniformly bounded. Independent phases on incidence components in a local limit replace the symmetric graph hypothesis used in the fourth paper. The argument does not establish a finite-field or nonlinear theorem.

(3) **An all-code theorem, and the failure of its universal structural premise.** Uniform hyperfiniteness together with paired-distance separation gives a half-rate bound for every finite alphabet and every recovery function. A native expander-shear construction shows that actual cone incidence need not be hyperfinite after cheap genie deletion. Its own cone relations nevertheless satisfy an all-code half bound, by a different argument using physical row lanes.

(4) **Connected demands with a positive coding deficit.** A conjugated native construction keeps a positive fraction of matched ports connected after the prescribed deletions. A full-cone row analysis still yields an independent coordinate rectangle of positive density. The resulting nonlinear rate deficit is independent of the fixed depth parameter. It is a theorem about these relations and an explicitly stated row interface, not a normal form for arbitrary tape computations.

The appendix develops a finite-field matrix-label certificate. It supplies characteristic-two half-rate bounds at arbitrary vector width on a separate expanding family constructed to admit the certificate. It neither realizes that family by a tape schedule nor transfers the certificate to every paired-distance-separated system.

The main statements are Theorem 3.3 (the native structural bridge), Theorem 4.5 and Corollary 4.7 (complex linear codes), Corollary 4.4 (all codes under hyperfiniteness), Corollary 5.9 (native expansion with nonlinear half bounds), and Theorems 6.3 and 6.8 (connected demands with a depth-independent coding deficit). The abstract finite-field family is Theorem A.3.

1.3 Relation to earlier results

The first paper's causal block-network formulation and common-monotone-pass bounds remain distinct from the step-level incidence approach [5]. The second paper shows why data-independent geometric conditions and cheap encoders alone do not force the desired coding restrictions [7]. The bit-to-word reduction of the third paper concerns the computational task rather than the graph hypothesis studied here [6]. The fourth paper supplies the depth-to-index-code decomposition, home-based depth reduction, and a symmetric analytic half-rate theorem [4].

Storage codes, guessing games, and index coding have an established duality theory [8, 2]. We

use its elementary fibre implication, proved below, and do not need an asymptotic duality theorem to convert a storage bound into a genie lower bound. The analytic proof uses the local-limit and finite-trace framework for unimodular networks [1]. The new issue is the directed incidence geometry and its connection to actual input and output ports.

Two distinctions govern the conclusions. First, the native constructions are inexpensive executors, not transposition algorithms; their role is to test proposed universal graph properties. Second, although their coding bounds exclude transposition on the particular schedules, that algorithmic exclusion is already available from the first paper's pass machinery. The contribution here is the all-code statement for the full cone relation, uniformly over the specified genies and coordinate restrictions.

1.4 Quantifier conventions

Unless otherwise stated, machine resources, time constants, cone depth, incidence degree, and vector width are fixed before d or the number of coordinates tends to infinity. An error $o_D(L)$ may have a convergence threshold depending on the fixed depth D. A rate deficit described as independent of D has one positive leading constant valid for every fixed D; this does not make its error bound uniform when D grows. All nonlinear code bounds quantify over every finite alphabet and every storage code on the stated relation. No numerical experiment is a premise of a theorem.

Reading the proofs. The conditioning, fibre, projection, rectangle, threshold-pullback, port-color, component, separator, damaged-expander, rank-identity, matrix-label, and hyperfinite half-rate results are formalized in Lean. The paired-distance bound, self-demands, degree trimming, the depth-independent code deficit, and the expanding-family linear-code corollary are partially formalized. The remaining native-computation and complex-analytic arguments have written proofs and are not yet formalized. Imported results state their source and verification scope. The declaration index and verification appendix record formal counterparts as they become available.

2 Recovery relations and the cost of conditioning

Write $[m] = \{0, \dots, m-1\}$. For a matrix of entry width b, identify its raw coordinates with

$$V = [d] \times [d] \times [b], \qquad \mathrm{addr}(i,j,s) = (id+j)b + s, \qquad \Pi(i,j,s) = (j,i,s).$$

Thus Π is an involution, and the raw bit at p is demanded at physical output address $\mathrm{addr}(\Pi(p))$. In particular, for $p = (i,j,s)$ its raw row is i and its physical output row is j. These two row coordinates must remain distinct in arguments about actual tape geometry. We sometimes write $\Pi(p)$ for the corresponding output address when the address identification is understood.

Definition 2.1 (Storage code)**.** Let V be a finite coordinate set and $N(q) \subseteq V$ a recovery set for every $q \in V$. A storage code over a finite alphabet $\mathcal{A}$ is a set $\mathcal{C} \subseteq \mathcal{A}^V$ such that

$$x, y \in \mathcal{C}, \quad x|_{N(q)} = y|_{N(q)} \quad \Longrightarrow \quad x_q = y_q.$$

Equivalently, each coordinate is a function of its recovery set on $\mathcal{C}$. The functions may be nonlinear and may depend on the code. For $a = |\mathcal{A}| \geq 2$, the rate of a nonempty code is $|V|^{-1} \log_a |\mathcal{C}|$.

The induced relation on $Q \subseteq V$ is $N_Q(q) = N(q) \cap Q$ for $q \in Q$. A code on this induced relation need not extend to one on V. Conversely, projecting an arbitrary code on V to Q need not give a code for N_Q: the omitted coordinates must first be fixed.

Lemma 2.2 (Paid conditioning)**.** Lean: `isStorageCode_conditionedCode`, `storageCode_conditioning_bound` [`StorageCode.lean`]

Let $H = V \setminus Q$. Every nonempty fibre of a storage code $\mathcal{C}$ obtained by fixing the coordinates in H, and then restricting to Q, is a storage code for N_Q. If every code for N_Q has at most a^u words, then

$$|\mathcal{C}| \le a^{|H|+u}.$$

Outline of the formal proof. Two words in a fixed fibre agree on H. Agreement on $N(q) \cap Q$ therefore gives agreement on all of $N(q)$, so it determines q. There are at most $a^{|H|}$ fibres, each of size at most a^u. □

Definition 2.3 (Paired incidence)**.** The incidence graph $E(N)$ has vertex set $\{L_p : p \in V\} \sqcup \{R_q : q \in V\}$, with an edge L_pR_q exactly when $p \in N(q)$. The pair L_p, R_p is the *demand pair* for p; it is not an extra graph edge. Let

$$b_E(r) = \big|\{p \in V : \mathrm{dist}_E(L_p, R_p) \le r\}\big|,$$

where disconnected vertices have distance ∞. A sequence on $|V| \to \infty$ satisfies the *paired-distance condition* if $b_E(r) = o(|V|)$ for every fixed integer r.

The two incidence degrees encode different restrictions: $\deg(R_q) = |N(q)|$ bounds recovery-set size, whereas $\deg(L_p)$ bounds the reuse of a coordinate across recovery sets. Bounded cone depth immediately gives the former, not the latter. If N is symmetric, the incidence graph is the bipartite double cover of its coordinate graph. In that special case a path from L_p to R_p projects to an odd closed walk based at p. For directed recovery, forgetting which endpoint supplies information loses this interpretation.

Definition 2.4 (Index code)**.** For binary inputs on V, an index code of length m for N is a map $G : \{0,1\}^V \to \{0,1\}^m$ such that $G(x)$ together with $x|_{N(q)}$ determines x_q for every q.

Lemma 2.5 (Fibre bound)**.** Lean: `isStorageCode_indexFiber`, `indexCode_length_ge` [`StorageCode.lean`]

Every fibre of an index code for N is a binary storage code for N. Consequently, if all such storage codes have size at most $2^{|V|-s}$, every index code has length at least s.

Outline of the formal proof. Inside a fibre the broadcast is fixed, so agreement on the recovery set determines the demanded bit. The at most 2^m fibres partition $2^{|V|}$ inputs. Hence $2^{|V|} \le 2^m 2^{|V|-s}$. □

For a native computation, let $I_\Gamma(\ell)$ be the raw input coordinates in the dependency cone of physical output ℓ after a genie Γ is fixed. The relation relevant to transposition is

$$N_\Gamma(q) = I_\Gamma(\Pi(q)). \tag{2.1}$$

If the machine correctly transposes every input and the schedule is oblivious, its revealed genie word is an index code for this one relation. If the whole-step value width is β, that word has at most $\beta|\Gamma|$ bits. Thus a storage bound of $2^{(1-\rho)L+o(L)}$ contradicts a genie of cost at most εL whenever $\varepsilon < \rho$. The schedule may be studied without any correctness assumption; correctness is used only in this last identification of the demanded output with the raw bit q.

All entropies below are Shannon entropies in bits unless a base is displayed. For a random word supported on a storage code, every recovery relation holds almost surely. Deterministic projection arguments therefore bound arbitrary such distributions as well as uniform distributions on the code. We state cardinality bounds explicitly so that no support-size conclusion is inferred merely from a low entropy under a particular input law.

3 Paired distances in actual tape cones

The recovery relation extracted from a computation is directed: an input coordinate may occur in the cone of an output without the converse occurrence. The appropriate geometric object retains separate copies of the input and output ports. We prove that, after a paid bound on input reuse, almost every matched pair of ports is far apart in this incidence graph. No assumption that the computation transposes is needed.

3.1 The computation DAG and the incidence graph

Fix integers $d \geq 2$ and $b \geq 1$, and write

$$L = d^2 b, \qquad W = db, \qquad [L] = \{0, \ldots, L-1\}.$$

Identify a coordinate (i, j, s), with $0 \leq i, j < d$ and $0 \leq s < b$, with the raw address $(id + j)b + s$. The transpose permutation is

$$\Pi((id + j)b + s) = (jd + i)b + s. \tag{3.1}$$

Throughout this section, logarithms and entropies have base two.

Consider an execution with one read-only input tape, $k \geq 1$ work tapes, and exactly L output appends in T steps. Each head starts at zero and moves by at most one cell per step. At most one bit is appended per step. Write $x_0(t)$ for the input-head position and $x_j(t)$ for work-head position j immediately before step t. A work cell is written at every visit, including a visit which preserves its symbol. Let

$$\mathrm{prev}_j(t) = \max\{u < t : x_j(u) = x_j(t)\}$$

when this set is nonempty. We first take the schedule to be oblivious: these positions, the append times, and T depend on the input length alone. The structural argument will also apply to a single frozen adaptive trace.

Definition 3.1 (Whole-step cones)**.** The value v_t of step t consists of the state after that step, all symbols written on the k work tapes, and the appended output bit or a symbol indicating no append. Fix a constant integer β large enough to encode one such value. A genie $\Gamma \subseteq [0, T)$ reveals $(v_t)_{t \in \Gamma}$ and has cost $\beta|\Gamma|$ bits.

The computation DAG H has vertices $0, \ldots, T-1$, state edges $t \to t-1$ for $1 \leq t < T$, and cell edges $t \to \mathrm{prev}_j(t)$ whenever the displayed predecessor exists. Parallel edges may be identified. All edges point backward in time.

A path $t_0 \to \cdots \to t_m$ is Γ-free if $t_1, \ldots, t_m \notin \Gamma$; its first vertex may belong to Γ. The genie has *full depth at most* D if every such path has at most D edges. Let $C_\Gamma(t)$ be the set of endpoints of Γ-free paths from t. If output address ℓ is appended at step t_ℓ, put

$$I_\Gamma(\ell) = \{x_0(u) : u \in C_\Gamma(t_\ell)\} \cap [L], \qquad N(q) = I_\Gamma(\Pi(q)).$$

These are the whole-step conventions of [4, the dependency DAG and cone lemma]. In particular, a length-zero path is permitted, even when its vertex is revealed. Here $N(q)$ is a syntactic input set; its definition does not require that output $\Pi(q)$ equal input q.

Each vertex of H has both in-degree and out-degree at most $k + 1$. For the in-degree assertion, a given visit has at most one next visit to the same cell on each work tape, and at most one next state edge. Consequently, if the genie has full depth at most D, then

$$|N(q)| \leq B, \qquad B = \sum_{h=0}^{D} (k+1)^h. \tag{3.2}$$

Definition 3.2 (Paired incidence graph)**.** The graph E has distinct vertices L_p, R_p for every $p \in [L]$, with an undirected edge L_pR_q precisely when $p \in N(q)$. The physical port represented by L_p is input address p; that represented by R_q is output address $\Pi(q)$. For $Q \subseteq [L]$, let E_Q be the induced graph on $\{L_p, R_p : p \in Q\}$. Distances between vertices in different components are infinite.

The graph E is an auxiliary bipartite graph on physical ports. It is neither the computation DAG H nor the undirected graph obtained by forgetting the directions of the recovery relation on $[L]$. Its bipartiteness is automatic; the assertion below concerns the distances between the two copies of the *same* coordinate.

Theorem 3.3 (Paired-distance bound)**.** Lean (partial): `short_paired_count_log_bound`, `short_paired_count_log_asymptotics` [`PathPackingAsymptotics.lean`]
Not formalized: The native realization and disjoint-path transport estimates, including their constants uniform in the schedule, genie, and entry width.

Fix $k \geq 1$, $D \geq 0$, $c > 0$, a degree bound Δ, and an integer $r \geq 1$. Suppose $T \leq c(L+1)$ and Γ has full depth at most D. For every $Q \subseteq [L]$ such that E_Q has maximum degree at most Δ,

$$\left|\{p \in Q : \operatorname{dist}_{E_Q}(L_p, R_p) \leq r\}\right| = O_{k,D,c,\Delta,r}\left(\frac{L}{\log_2 d}\right). \tag{3.3}$$

The constant is uniform over $b \geq 1$, the schedule, the genie, and Q. Neither correctness as a transposer nor a bound on $|\Gamma|$ is assumed.

The proof uses a geometric relaxation in which selected paths may share a location and subsequently separate. These paths are auxiliary objects; they do not assert that a coded tape symbol carries independent source bits. A row-entropy potential controls partial transpose transport in this relaxation. Alternating forward and reverse realizations of cone matchings then converts short paired paths in E_Q into that transport.

3.2 A partial-transport inequality

We specify the geometric relaxation so that its use does not import a routing assumption about the original machine. There are K tapes, one unit-speed head on each, and ρ registers without spatial addresses. A geometric tick first permits arbitrary transfers among the current head cells and registers, leaving other locations unchanged, and then moves each head by at most one cell. Each selected source label follows a single trajectory. Different trajectories may occupy the same slot and later separate. The *occupancy* is the number of selected trajectories at a slot, without regard to encoded values. We bound occupancy at each placement before and after a local transfer. Moving heads does not change the locations of the selected labels.

Lemma 3.4 (Partial transpose transport)**.** Not formalized; a complete proof is given below.

Let $S \subseteq [L]$ have $m > 0$ elements. Suppose geometric trajectories start at the distinct input cells $p \in S$, with no selected labels elsewhere, and end at the distinct cells $\Pi(p)$ of one output tape. Assume all heads start at zero, every slot has occupancy at most C, and the trajectories use U geometric ticks. Then

$$m \log_2 d - m \log_2\left(\frac{L+2W}{m}\right) - m \leq CK(K+\rho+3)U. \tag{3.4}$$

In particular, including $m = 0$,

$$m \log_2 d \leq CK(K+\rho+3)U + 3L. \tag{3.5}$$

Proof (not formalized). Fix all trajectories before choosing any partition. For each $\theta \in \{0, \dots, W-1\}$, partition every tape into the bins $\lfloor (z-\theta)/W \rfloor$, retaining the tape name as part of the bin name. Merge the K bins currently containing heads, together with all registers, into one active class. The remaining bins are separate classes. The active class contains at most $KW + \rho$ slots.

Choose a source label V uniformly from S, and let $A = \lfloor V/W \rfloor$ be its original input row. If $Z_{t,\theta}$ is the class containing its trajectory at boundary t, define

$$\Phi_t = m\, \mathbb{E}_\theta \mathrm{H}(A \mid Z_{t,\theta}).$$

This is entropy of a random *source label*, not entropy of the machine input or the contents of a cell. For any two class variables Z, Z', the chain rule gives

$$\mathrm{H}(A \mid Z') - \mathrm{H}(A \mid Z) \le \mathrm{H}(Z \mid Z'). \tag{3.6}$$

Thus a new class containing n labels contributes at most n times the entropy of their old class names to the increase of the unnormalized potential.

A local transfer changes no label's class: both of its endpoints lie in the active class. Analyze a simultaneous movement of the heads as K successive movements, with all labels held at their physical slots. A unit movement of a particular head crosses a bin boundary for exactly one offset θ. At that offset, the new active class has at most two old class names, namely the old active class and the newly joined bin. A released bin has only the old active name. All other classes are unchanged. By occupancy, the new active class contains at most $C(KW + \rho) \le C(K + \rho + 3)W$ selected labels. Equation (3.6) therefore bounds the increase, after averaging over θ, by $C(K + \rho + 3)$. A pause has zero cost. Summing gives

$$\Phi_U - \Phi_0 \le CK(K + \rho + 3)U. \tag{3.7}$$

Initially, a shifted input bin meets at most two aligned input rows. Only the input tape contains selected labels, and merging headed bins merges no additional occupied input bin. Hence $\Phi_0 \le m$.

For the endpoint estimate, first keep all output bins separate. At output position $v = \Pi(p)$ the original source row is $\lfloor (v \bmod W)/b \rfloor$. This label is W-periodic, with exactly b occurrences of each row in a period. Thus a shifted W-cell bin containing n selected labels has row multiplicities $n_a \le b$, and

$$n\, \mathrm{H}(A \mid \text{this bin}) = n \log_2 n - \sum_a n_a \log_2 n_a \ge n \log_2 n - n \log_2 b.$$

Use $0 \log_2 0 = 0$. At most $L/W + 2$ shifted bins meet $[L]$. Convexity of $x \log_2 x$ consequently gives

$$m\, \mathrm{H}(A \mid \text{separate output bins}) \ge m \log_2 \left(\frac{m}{b(L/W + 2)} \right) = m \log_2 d - m \log_2 \left(\frac{L + 2W}{m} \right).$$

Merging headed bins is a coarsening of the partition, so it cannot decrease this conditional entropy. This is a lower bound for Φ_U. Together with (3.7) and $\Phi_0 \le m$, it proves (3.4).

For $a > 0$ and $0 < x \le a$, $x \log_2(a/x) \le a/(e \ln 2)$. Since $L + 2W \le 2L$ and $m \le L$,

$$m + m \log_2 \left(\frac{L + 2W}{m} \right) \le \left(1 + \frac{2}{e \ln 2} \right) L < 3L.$$

Rearranging proves (3.5). □

The factor b in the endpoint estimate is essential. The statement holds for arbitrary selected source subsets and every entry width; no conversion of a word instance into a bit instance is used.

3.3 Realizing and reversing incidence matchings

Lemma 3.5 (A matching of cone incidences)**.** Not formalized; a complete proof is given below.

Let $\mathcal{M}$ be a matching in E. There is a geometric realization of its edges, carrying a label from input cell p to output cell $\Pi(q)$ for each $L_pR_q \in \mathcal{M}$, with $K = k+2$ tapes, one register, occupancy at most $B+1$, and T ticks. All such realizations use the same head configurations h_0 at entry and h_T at exit. The geometric paths can be reversed in $2T$ ticks with the same occupancy, from configuration h_T to h_0.

Proof (not formalized). For every edge L_pR_q choose one Γ-free backward path from the output step $t_{\Pi(q)}$ to a step reading raw input p. It has at most D edges. Any vertex of the computation DAG belongs to at most B chosen paths: the distinct output steps of paths using that vertex lie within its forward radius D, which contains at most B vertices. This counts output endpoints, not the number of possible combinatorial paths.

For the geometric realization, regard the input tape and the append-output tape as two additional work-capable tapes. The input head follows its original trajectory. The output head occupies the next append address and advances exactly when an append occurs. The additional register represents a whole step's record for purposes of the geometric interface; it is not restricted to the original control-state component.

Read each chosen DAG path in forward chronological order. Its label waits at its raw input cell until the chosen input read. A state edge is realized by waiting in the record register from one step to the next. A cell edge is realized by waiting in that work cell between two consecutive visits. At a path vertex, the incoming head cell or register and the required outgoing head cell or register are present at the local interface, so the geometric model permits the transfer. At the chosen output step the label moves to the current output cell and remains there. These operations occur before the original head movements, and therefore use the original T ticks.

During a work-cell waiting interval all labels there use its same previous-visit vertex, so there are at most B. Labels in the record register use one step vertex and satisfy the same bound. A raw input cell contains only its own not-yet-departed source label; a completed output cell contains only its own output label. The matching supplies distinct sources and outputs, and the input and output tapes are distinct. The bound $B+1$ is thus conservative at every placement. At the end of the block, all selected labels occupy output ports and none remains in a work cell or register.

The record register is an augmentation of the geometric model. A syntactic state edge of the whole-step DAG may refer to components that the real control state does not retain. The argument does not identify that edge with an independent information channel in the original machine. Instead it uses an additional register in precisely the geometric model covered by Lemma 3.4.

To reverse a tick, first preserve all labels and undo the original unit head movements in one tick. In a second tick, keep the heads fixed and reverse the selected local transfers. Both endpoints of each transfer are again current head cells or registers. This reverses the geometric trajectories, including waiting intervals, with the same occupancy. It does not invert a transition function or reconstruct a past machine state. All matchings use the same original head schedule, including the two port tapes. Their forward and reverse configurations therefore agree exactly at the stated block boundaries. □

Lemma 3.6 (Disjoint short paired paths)**.** Not formalized; a complete proof is given below.

For any fixed odd $s \geq 1$, the number m of vertex-disjoint s-edge paths in E whose endpoints are L_p, R_p satisfies

$$m \leq A_s \frac{L}{\log_2 d}, \qquad A_s = 4sc(B+1)(k+2)(k+6)+3. \tag{3.8}$$

Proof (not formalized). Write $s = 2a + 1$ and orient each incidence path from its left endpoint to its right endpoint. The edges at a fixed position along these paths form a matching, because the paths are vertex-disjoint. Realize the first matching forward, the second backward, and continue alternating, ending with a forward block.

For example, an incidence path beginning L_p, R_v, L_u, R_w first carries its label from input p to output $\Pi(v)$, then from that same output cell back to input u, and then to output $\Pi(w)$. Every splice identifies the same physical tape and address. In particular, Π is never performed as a free operation; it only names which output cell a right port represents. Forward blocks end at h_T, where reverse blocks begin, and reverse blocks end at h_0, where the next forward blocks begin. No head repositioning is omitted. At each junction the selected labels occupy distinct ports, so occupancy does not accumulate across stages.

The resulting trajectories carry each original input p to output $\Pi(p)$, starting with every head at zero. They use $K = k + 2$ tapes, $\rho = 1$, occupancy $B + 1$, and

$$U = (3a + 1)T \le 2sT \le 4scL$$

ticks. Apply (3.5) to obtain (3.8). □

Proof of Theorem 3.3. For each $p \in Q$ at paired distance at most r, choose one shortest path in E_Q. Its length is odd. Put

$$V_r = \sum_{j=0}^{r} \max(1, \Delta)^j.$$

A fixed vertex z belongs to at most V_r chosen paths: the distinct left endpoints of paths through z are within distance r of z, and there is one chosen path per left endpoint.

For each odd $s \le r$, choose a maximal vertex-disjoint subfamily among the chosen paths of length s. Every candidate meets a selected path. A selected path has $s + 1$ vertices and therefore meets at most $(s + 1)V_r$ candidates. Lemma 3.6 gives the explicit bound

$$\left|\{p \in Q : \mathrm{dist}_{E_Q}(L_p, R_p) \le r\}\right| \le \sum_{\substack{1 \le s \le r \\ s \text{ odd}}} (s + 1) V_r A_s \frac{L}{\log_2 d}. \tag{3.9}$$

All parameters in the sum are fixed independently of b, the schedule, and the genie. This proves the theorem. □

3.4 Input reuse, conditioning, and adaptive traces

Corollary 3.7 (Self-demands)**.** Lean (partial):
`self_demands_count_of_disjoint_transport` [`SelfDemands.lean`]
Not formalized: The native one-edge transport estimate.

Without any bound on left incidence degrees,

$$|\{p \in [L] : p \in N(p)\}| \le A_1 \frac{L}{\log_2 d}.$$

Proof (not formalized). The edges $L_p R_p$ for self-demanding coordinates form a matching. Apply Lemma 3.6 with $s = 1$. □

Corollary 3.8 (Paid degree reduction)**.** Lean (partial):
`paid_degree_trimming` [`IncidenceTrimming.lean`]
Not formalized: The native cone-size premise and the subsequent paired-distance asymptotic application.

For an integer $M \geq 1$, let

$$H_M = \{p \in [L] : \deg_E(L_p) > M\}, \qquad Q_M = [L] \setminus H_M.$$

Then $|H_M| \leq BL/M$, and E_{Q_M} has maximum degree at most $\max(B, M)$. For each fixed $\delta \in (0, 1)$, choosing the fixed threshold $M \geq B/\delta$ retains at least $(1 - \delta)L$ coordinates and gives (3.3) at every fixed radius.

Proof (not formalized). By (3.2), $\sum_p \deg_E(L_p) = \sum_q |N(q)| \leq BL$. This bounds the number of left degrees exceeding M. Deleting both ports of each such coordinate cannot increase a degree. Surviving left degrees are at most M, and all right degrees were at most B. □

The deletion in Corollary 3.8 costs a small *fixed* fraction for a fixed threshold; it is not an uncharged $o(L)$ deletion. Its coding interpretation is paid conditioning as in Lemma 2.2. Proposition 8.2 below spells out the resulting conditional transfer of an all-code half bound.

Remark 3.9 (Exactly what is structural)**.** The premise of Proposition 8.2 is an additional coding statement. Theorem 3.3 alone supplies no code-rate bound. It also supplies no genie: the depth D is assumed fixed before d tends to infinity. Its constants may grow rapidly with D.

Every argument proving Theorem 3.3 applies to one fixed legal adaptive execution trace with the stated time, output, and depth properties. Freeze that trace's head walks and append times and use them as the geometric schedule. The only random object in Lemma 3.4 is an auxiliary source label, so no distribution of inputs realizing the trace is required. This observation does not produce a common cone relation across inputs, bound the entropy of a trace, or make the input coordinates independent on a trace fibre. Those are separate obligations for an adaptive all-input coding reduction.

3.5 Why symmetrizing the recovery relation loses the conclusion

The distinction between incidence distance and ordinary odd cycles is necessary even for simple native operations. The example below forces directed triangles in every cone relation while respecting the geometric conditions already proved.

Proposition 3.10 (Persistent directed triangles)**.** Not formalized; a complete proof is given below.

For canonical widths $b = \max(1, \lceil \log_2 d \rceil)$, there is a fixed uniform deterministic oblivious native machine with $O(L)$ runtime which appends exactly L bits and computes

$$Y[j, i, s] = X[j, -j - i \bmod d, s].$$

For every genie Γ, its demand-labelled cone relation contains

$$F(i, j, s) = (j, -i - j \bmod d, s) \in N(i, j, s). \tag{3.10}$$

The map F has $\gcd(3, d)b$ fixed points and otherwise consists of disjoint directed three-cycles. Thus the undirected symmetrization of the recovery relation contains $(L - \gcd(3, d)b)/3$ vertex-disjoint triangles. Making it triangle-free by coordinate deletion requires at least this many deletions.

Moreover, for each fixed $\varepsilon > 0$, the machine admits genies of cost at most εL and full depth bounded independently of d. The same triangle conclusion holds for those genies.

Proof (not formalized). Output rows j in order and, within each row, output entries i in order. Obtain the entry directly from input row j, column $c = -j - i \bmod d$. Read its b bits forward, appending each bit in the same transition that reads it. Between successive entries the target column decreases by one. An ordinary repositioning costs $O(b)$ input-head steps, and the single wrap within a row costs $O(db)$. Moving from the final target of one row to the initial target of the next costs $O(db)$. The total input motion and emission cost is therefore $O(d^2 b)$.

Here is a fixed-resource implementation of the indexing and motion control. First count the raw length L with a little-endian binary counter, returning its head to zero after each increment. The total carry distance is $O(L)$. Rewind the input head, and recover d, b from the strictly increasing function $d \mapsto d^2 \max(1, \lceil \log_2 d \rceil)$ by binary search and ordinary arithmetic on $O(\log L)$ bits. This setup takes $\mathrm{poly}(\log L) = O(L+1)$ time on a fixed number of tapes. Noncanonical lengths may use an immediate fallback after this test.

At canonical lengths $d \geq 2$, every subsequent index and displacement has $O(b)$ bits. Per-entry operations are addition, subtraction, comparison, and modular decrement, implementable in $O(b)$ time by full scans of a fixed number of binary registers. A reusable unary guide of length b controls blocks of b input moves or output appends. Each use scans outward and returns, with the input head paused during the return. Longer displacements use $O(d)$ such blocks with $O(b)$ counter work per block; they cost $O(db)$ each. Hence all entry operations, row wraps, and row transitions still cost $O(L)$. The work tapes are registers or guides accessed by monotone outward-and-return scans from zero, so their walks are home-based. All motion depends on the public length and counters; payload bits affect only the emitted values. This proves uniformity, obliviousness, and the claimed native cost with a fixed tape alphabet and control.

The step emitting physical output (j, i, s) reads raw input $(j, -j - i \bmod d, s)$. The length-zero path at this step is Γ-free for every Γ, including one revealing that step. Demand relabelling by (3.1) proves (3.10). Direct calculation gives

$$F^2(i, j, s) = (-i - j, i, s), \qquad F^3(i, j, s) = (i, j, s).$$

Its fixed points satisfy $i = j$ and $3i = 0$ in $\mathbb{Z}/d\mathbb{Z}$, so there are exactly $\gcd(3, d)b$ of them. All other orbits have size three and give vertex-disjoint triangles after symmetrization.

Finally use the home-based depth-reduction theorem and long-edge conversion of [4, Theorem B and the proof of Corollary B]. For fixed machine resources and each $\varepsilon > 0$, these give a genie with cost at most εL and constant full depth. That construction uses the schedule and its runtime, not transpose correctness. It therefore applies to this machine. The triangles persist because (3.10) held for every genie. □

The guaranteed arcs in this example contribute only the incidence matching $L_{F(q)}R_q$; they do not by themselves connect L_q to R_q around the directed three-cycle. There is therefore no conflict with Theorem 3.3. Additional actual cone edges are permitted throughout the argument. The example rules out a universal local bipartiteness conclusion for the symmetrized recovery graph, while leaving the directed paired-distance condition intact.

4 Two half-rate bounds for directed recovery

The distinction between a coordinate and its two incidence ports is essential here. A directed recovery relation on a finite set V, with $N = |V|$, assigns a set $\mathcal{N}(q) \subseteq V \setminus \{q\}$ to each $q \in V$. Its *paired incidence graph* E has ports L_p, R_p, $p \in V$, and an undirected edge $L_p R_q$ precisely when $p \in \mathcal{N}(q)$. Distances below are in this graph; disconnected ports have distance infinity. Write

$$b_E(r) = \big|\{p \in V : \mathrm{dist}_E(L_p, R_p) \leq r\}\big|. \tag{4.1}$$

A sequence (E_n) on $2N_n$ ports, with $N_n \to \infty$, has *paired-distance separation* if $b_{E_n}(r) = o(N_n)$ for every fixed integer $r \geq 1$. This concerns prescribed pairs of ports, rather than the girth or connected-component sizes of E_n.

We prove a half-rate bound under either of two additional hypotheses. Uniform hyperfiniteness suffices for every finite-alphabet code. Without hyperfiniteness, bounded complex linear recovery coefficients and a fixed block width suffice. The latter extends the graph hypothesis of Theorem A of [4] from a locally bipartite underlying graph to paired-distance separation.

4.1 All codes on hyperfinite incidence graphs

A storage code over a finite alphabet $\mathcal{A}$ of size at least two is a set $\mathcal{C} \subseteq \mathcal{A}^V$ such that two codewords agreeing on $\mathcal{N}(q)$ agree at q, for every $q \in V$. Equivalently, each coordinate has a recovery function on the neighbour assignments that occur in the code. The code need not be linear, and no distribution or extension property is assumed.

Lemma 4.1 (Separated components)**.** Lean:
`storageCode_card_le_of_separated_components` [`IncidenceComponents.lean`]

If L_p and R_p lie in different components of E for every $p \in V$, then every storage code satisfies

$$|\mathcal{C}| \leq |\mathcal{A}|^{\lfloor N/2 \rfloor}.$$

Outline of the formal proof. Let $\alpha(p)$ and $\beta(p)$ be the components containing L_p and R_p. Choose an order of the components, and record the coordinates p for which $\alpha(p)$ precedes $\beta(p)$. This record determines a codeword. Indeed, process the α-components in increasing order. A coordinate in the current component was either recorded, or its β-component is earlier. In the latter case every recovery neighbour $u \in \mathcal{N}(p)$ satisfies $\alpha(u) = \beta(p)$, so its value is already determined. The equality case $\alpha(p) = \beta(p)$ is excluded by hypothesis. Compare this order with its reverse. Every coordinate is recorded in exactly one of the two orders, so one record has at most $\lfloor N/2 \rfloor$ coordinates. Its projection is injective on $\mathcal{C}$. □

Theorem 4.2 (Finite separator bound)**.** Lean:
`storageCode_separator_log_bound` [`IncidenceSeparator.lean`]

Let Z be a set of incidence ports such that each component of $E - Z$ has at most K vertices, where $K \geq 1$. Every nonempty storage code satisfies

$$\log_{|\mathcal{A}|} |\mathcal{C}| \leq \frac{N + |Z| + b_E(K)}{2}. \tag{4.2}$$

No degree bound is required.

Outline of the formal proof. Let U contain the coordinates with at least one port in Z, and also the coordinates counted by $b_E(K)$. Thus $|U| \leq |Z| + b_E(K)$. Delete both ports of every coordinate in U, obtaining the induced relation on $Q = V \setminus U$. Its components still have at most K ports. If a surviving pair were connected, a shortest path in its component would have length at most $K - 1$, contradicting the choice of U.

Fix an assignment to the coordinates in U. The restrictions to Q of codewords with this assignment form a storage code for the induced relation: deleted recovery arguments are fixed constants. Lemma 4.1 bounds each such fibre by $|\mathcal{A}|^{|Q|/2}$. There are at most $|\mathcal{A}|^{|U|}$ fibres, giving $|\mathcal{C}| \leq |\mathcal{A}|^{|U|+|Q|/2}$ and the claimed inequality. This conditions on coordinate values; it does not discard side-information edges without paying for their coordinates. □

Definition 4.3. Lean: `UniformlyHyperfinite` [`Hyperfinite.lean`]

A sequence (E_n) is *uniformly hyperfinite* if, for each $\eta > 0$, there is an integer K_η such that, for every sufficiently large n, one can delete at most $2\eta N_n$ incidence ports and leave components of size at most K_η.

Corollary 4.4 (Hyperfinite half-rate bound)**.** Lean: `hyperfinite_rate_limsup_le_half` [`Hyperfinite.lean`]

Suppose (E_n) is uniformly hyperfinite and has paired-distance separation. For any finite alphabets $\mathcal{A}_n$ of size at least two, and any nonempty storage codes $\mathcal{C}_n$ on these relations,

$$\limsup_{n\to\infty} \frac{\log_{|\mathcal{A}_n|} |\mathcal{C}_n|}{N_n} \le \frac{1}{2}.$$

Outline of the formal proof. For fixed η, Theorem 4.2 gives a limiting upper bound $1/2+\eta$, because K_η is fixed before n tends to infinity. Then let η decrease to zero. The finite estimate is uniform in the alphabet, so it also permits the displayed varying alphabets. □

4.2 Bounded complex linear recovery

Let $t \ge 1$. A recovery matrix A has $t \times t$ blocks A_{qp}, with $A_{qq} = 0$ and $A_{qp} = 0$ unless $L_p R_q \in E$. Its actual support graph has $L_p R_q$ precisely when $A_{qp} \neq 0$; we may use a larger declared relation E. A linear storage space with these recovery rules is contained in $\ker(I - A)$. No symmetry or adjoint relation between A_{qp} and A_{pq} is imposed.

Theorem 4.5 (Directed complex half-rank bound)**.** Not formalized; a complete proof is given below.

Fix integers $\Delta, t \ge 1$ and a real number $B \ge 1$. Let (E_n) have maximum incidence degree at most Δ and paired-port separation. Let A_n be complex recovery matrices of block width t on E_n, with every scalar entry of modulus at most B. Then

$$\liminf_{n\to\infty} \frac{\operatorname{rank}_{\mathbb{C}}(I - A_n)}{tN_n} \ge \frac{1}{2}. \tag{4.3}$$

Consequently every associated complex or real linear storage space has dimension at most $(1/2 + o(1))tN_n$.

We give the analytic details because two points matter: a component marking must preserve unimodularity, and finite nullity passes to the limit in only one direction. We use the marked-network framework of Aldous and Lyons [1, Sections 2 and 5]. Uniformly rooted finite graphs of bounded degree, with marks in fixed compact spaces, have locally convergent subsequences; their limits satisfy the mass-transport identity. For a unimodular law μ, bounded measurable operator families equivariant under network isomorphisms, and independent of the distinguished root, form a von Neumann algebra. In its $t \times t$ matrix amplification the normalized trace is

$$\tau(T) = \mathbb{E}_\mu \frac{1}{t} \operatorname{tr}(T_{oo}), \qquad \tau(I) = 1. \tag{4.4}$$

It is tracial and positive. These are the same analytic inputs used in the proof of Theorem A in [4].

Proof of Theorem 4.5. Form the coordinate graph G_n by joining p and q if either directed incidence is present. Its degree is at most 2Δ. Record the two directed incidence flags as discrete marks, and the corresponding bounded coefficient blocks as compact marks. In particular, retain an incidence flag when its coefficient converges to zero. Choose a subsequence realizing the limsup of

the normalized nullities, then a locally convergent marked subsequence with unimodular limit μ. The root is uniform among coordinates, and the rooted limit retains the root's connected coordinate component. Write (G, A, o) for that limit. The row and column absolute sums of A are at most $\Delta t B$, so

$$\|A\| \le \Delta t B, \qquad \|I - A\| \le c := 1 + \Delta t B.$$

Separation in the limit. For fixed r, s, the event that a coordinate within G_n-distance s of the root has paired-port distance at most r has probability at most

$$\left(\sum_{j=0}^{s} (2\Delta)^j \right) \frac{b_{E_n}(r)}{N_n}.$$

It is determined by finitely many discrete incidence flags and therefore is a clopen local event. Its limiting probability is zero. Taking the countable union over r, s shows that, almost surely, *every* coordinate p of G has L_p, R_p in different components of the limit port graph E.

An invariant phase extension. Conditionally on this countable graph, give each component of E an independent uniform phase on the complex unit circle. This law can be specified without choosing an invariant enumeration: for a finite list of ports, give independent phases to its distinct component classes, and equal phases to ports in the same class. The cylinder laws are consistent. Connectivity of two ports is a countable union of finite-path events, so their component partition is Borel. The cylinder integrals, and then all bounded Borel integrals by a monotone-class argument, define a Borel probability kernel κ_G. Its law is unchanged by any isomorphism or change of root.

The resulting phase-marked extension $\widehat{\mu}$ is unimodular. Indeed, for any nonnegative doubly rooted test function f, average f over κ_G to obtain $\bar{f}$. The same kernel is used with either root distinguished. Applying mass transport to $\bar{f}$ and Tonelli's theorem gives

$$\mathbb{E}_{\widehat{\mu}} \sum_x f(G, o, x) = \mathbb{E}_{\mu} \sum_x \bar{f}(G, o, x) = \mathbb{E}_{\mu} \sum_x \bar{f}(G, x, o) = \mathbb{E}_{\widehat{\mu}} \sum_x f(G, x, o).$$

This construction is a probability-space extension, not an assertion that component phases are a factor of independent vertex labels.

Let D_L, D_R multiply the t entries at coordinate p by the phase of the component of L_p, R_p, respectively. They are measurable, equivariant diagonal unitaries. An edge $L_p R_q$ equates the phase at column p with that at row q, and hence

$$A D_L = D_R A. \tag{4.5}$$

The phases of each matched pair are independent and nonatomic. By countability they differ at every coordinate simultaneously, almost surely. Thus $D_L - D_R$ is injective; no bounded inverse is asserted.

The tracial half bound. Let $K = \ker(I - A)$ and let P_K be its orthogonal projection. If $y \in K$ and $D_L y \in K$, then $D_L y = A D_L y = D_R A y = D_R y$, so $y = 0$. Consequently $T = (I - P_K) D_L P_K$ has kernel $K^{\perp}$, and its range closure lies in $K^{\perp}$. All these operators lie in the amplified equivariant algebra. Polar decomposition there gives a partial isometry U with $U^*U = P_K$ and $UU^* \le I - P_K$. One can obtain U as the strong limit of $T(T^*T + \varepsilon I)^{-1/2}$ as $\varepsilon \downarrow 0$; functional calculus and strong closure keep this limit in the algebra. Therefore

$$\tau(P_K) = \tau(U^*U) = \tau(UU^*) \le 1 - \tau(P_K), \qquad \tau(P_K) \le \frac{1}{2}. \tag{4.6}$$

This uses the range closure, not an assumption that the range is closed. The expected root trace of P_K is unchanged on forgetting the phases.

Return to finite matrices. Put $M_n = I - A_n$, $H_n = M_n^* M_n$, and $H = (I-A)^*(I-A)$. Their normalized empirical spectral measures ν_n have common support in $[0, c^2]$. For each integer $j \geq 0$, the moment $(tN_n)^{-1} \operatorname{tr} H_n^j$ is the expected normalized root-block trace of a bounded continuous local function; radius $2j$ suffices. Local convergence gives moment convergence to the averaged root spectral measure ν of H. Polynomial approximation on $[0, c^2]$ gives weak convergence. The closed-set inequality for the singleton $\{0\}$ is

$$\limsup_n \frac{\dim \ker M_n}{tN_n} = \limsup_n \nu_n(\{0\}) \leq \nu(\{0\}) = \tau(P_K) \leq \frac{1}{2}. \tag{4.7}$$

Here $\ker(M^*M) = \ker M$ holds for finite matrices and bounded operators. This proves (4.3). Real matrices have the same rank after complexification. □

Corollary 4.6 (Uniform finite form)**.** Not formalized; a complete proof is given below.

For fixed Δ, t, B and every $\varepsilon > 0$ there exist an integer $r \geq 1$, a number $\delta > 0$, and N_0 such that

$$b_E(r) \leq \delta N, \quad N \geq N_0 \quad \Longrightarrow \quad \operatorname{rank}_{\mathbb{C}}(I - A) \geq (1/2 - \varepsilon) tN$$

for every relation and matrix with the bounds of Theorem 4.5.

Proof (not formalized). Otherwise choose a failure with $r = j$, $\delta = 1/j$, and $N \geq j$ for each positive integer j. These failures have paired-distance separation, contradicting Theorem 4.5. □

Corollary 4.7 (Complex linear codes on actual cone relations)**.** Not formalized; a complete proof is given below.

Fix the tape count, linear-time constant, full cone depth, block width t, and a uniform bound on the recovery coefficients. Every sequence of complex linear storage spaces on the corresponding native cone relations, admitting linear recovery maps with the stated block width and coefficient bound, has dimension at most $(1/2 + o(1))tL$ as $d \to \infty$.

Proof (not formalized). Remove the $O(L/\log d)$ self-demand coordinates supplied by Corollary 3.7. For any fixed $0 < \delta < 1$, also remove coordinates whose left degree exceeds a fixed $M \geq B_D/\delta$, where $B_D = \sum_{h=0}^{D} (k+1)^h$ is the cone bound in (3.2), using Corollary 3.8. Write H for their union and Q for the retained set. Then $|H| \leq \delta L + o(L)$, and the induced relation is loop-free, has bounded incidence degree, and satisfies the paired-distance condition by Theorem 3.3.

If $\mathcal{U}$ is the original linear space, its subspace with zero coordinates on H has codimension at most $t|H|$. On this subspace the original bounded linear recovery maps restrict to maps on Q with the same coefficient bound and block width. Theorem 4.5 bounds its dimension by $(1/2 + o(1))t|Q|$. Therefore

$$\dim \mathcal{U} \leq t|H| + (1/2 + o(1))t|Q| \leq (1/2 + \delta/2 + o(1))tL.$$

Take the size limit first and then let δ tend to zero. This is a complex-linear statement about the cone relations; it does not make the machine's binary genie a complex linear map. □

Remark 4.8 (Scope of the graph extension)**.** For a symmetric recovery relation, E is its bipartite double cover. A path between matched ports projects to an odd closed walk, and an odd cycle gives such a path at each of its vertices. With bounded degree, paired-distance separation is consequently equivalent to having only $o(N_n)$ vertices on odd cycles of every fixed bounded length. For directed relations the hypotheses differ: in disjoint directed seven-cycles with one recovery neighbour at each vertex, E is a union of isolated edges and every matched distance is infinite, whereas the underlying coordinate graphs have odd cycles at positive density. Thus replacing directed recovery by its undirected closure loses information.

4.3 What the complex bound says over finite fields

Corollary 4.9 (Necessary torsion in bounded lifts)**.** Not formalized; a complete proof is given below.

Fix a prime p, integers $\Delta, t \geq 1$, and $\varepsilon > 0$. Let (E_n) satisfy the graph hypotheses of Theorem 4.5. Suppose matrices M_n over $\mathbb{F}_p$ have diagonal blocks I_t, off-diagonal support in E_n, and $\dim \ker M_n \geq (1/2 + \varepsilon)tN_n$*. For any integer lifts $\widetilde{M}_n$ with the same support restrictions, literal diagonal blocks I_t, and uniformly bounded entries, at least $(\varepsilon - o(1))tN_n$ of their* nonzero *Smith invariant factors are divisible by p.*

Proof (not formalized). Set $\rho_n = \operatorname{rank}_{\mathbb{Q}} \widetilde{M}_n$. Theorem 4.5 gives $\rho_n \geq (1/2 - o(1))tN_n$. The unimodular transformations in Smith normal form remain invertible modulo p. Thus

$$\#\{\text{nonzero invariant factors divisible by } p\} = \rho_n - \operatorname{rank}_{\mathbb{F}_p} M_n \geq (\varepsilon - o(1))tN_n.$$

Equivalently, the subgroup annihilated by p in the integer cokernel has at least this dimension over $\mathbb{F}_p$. Zero invariant factors give free abelian summands and are not counted as torsion. □

This conclusion permits such torsion; it is not a finite-field half-rate theorem. A fixed extension field can be represented over its prime field by enlarging the fixed block width. The analytic argument does not cover growing block widths, unbounded complex coefficients, or nonlinear codes on nonhyperfinite relations. Appendix A gives a separate finite-field certificate on one existential expanding family, uniform in the storage-code block width. Neither result identifies all native cone relations with that family or settles the unrestricted coding problem.

5 A native nonhyperfinite family with a half-rate bound

The directed recovery formulation does not require a decomposition into small components. We demonstrate this with one fixed native computation. Its full cone incidence graphs contain robust expander cores, yet every storage code on every induced coordinate set has logarithmic cardinality at most half its coordinate count plus $o(L)$. This gives rate at most one half plus $o(1)$ whenever the retained set has size $\Omega(L)$. The computation is an invertible shear, not a transposition algorithm. The point of the example is the simultaneous graph-theoretic and coding conclusion for its actual execution.

Throughout this section let

$$d = 2m, \qquad b = \max\{1, \lceil \log_2 d \rceil\}, \qquad W = db, \qquad L = dW = d^2 b.$$

A raw coordinate is $p = (i, j, s) \in V = [d] \times [d] \times [b]$, where $[u] = \{0, \ldots, u-1\}$. Its physical address is $(id + j)b + s$, and its transpose demand is the physical output address of $\Pi(p) = (j, i, s)$. The paired incidence graph has distinct ports L_p, R_p; an edge $L_p R_q$ means that raw input p occurs in the full backward cone of the append at $\Pi(q)$. We write this graph as E_Γ when cones are stopped by a whole-step genie Γ. Deleting a coordinate always deletes both of its ports. In particular it is different from deleting chosen dependencies.

5.1 Two imported interfaces and a paid executor

We use the following expander input from the native shear construction of [7, section on expander shears and local recovery]. The graph construction is imported; the schedule refinements and the subsequent cone analysis will be given here.

Lemma 5.1 (Ordered expander input)**.** Cited: The expander-shear construction of [7]. The uniform list-generation refinement is justified below by a written argument; this full interface is not claimed to be Lean-verified.

There are fixed positive constants J, Δ, η, with J, Δ integers and $\eta \le 1$, and simple bipartite graphs H_m on

$$A_0, \dots, A_{m-1}, \quad B_0, \dots, B_{m-1},$$

such that their maximum degree is at most Δ, their edges partition into J partial increasing matchings, and

$$|N_{H_m}(S) \setminus S| \ge \eta |S| \quad \text{whenever } |S| \le m.$$

The matching endpoint lists can be generated uniformly in $O(m \operatorname{polylog} m)$ time using a fixed number of tapes and polylogarithmic working space besides the monotonically emitted lists. All constants and rational coefficients are fixed before m varies.

Here a partial increasing matching has its two endpoint sequences strictly increasing. The computational part of the imported interface can be seen directly from the rational overlap construction. Fix its finite family of increasing rational maps on interval domains. Their derivatives have fixed positive lower and upper bounds. An edge records a positive-length overlap of the image of one mesh cell with a second mesh cell; exact rational comparisons decide boundary coincidences. There are constantly many candidate neighbors in either direction. Split candidates by map name and by the residues of the two endpoint indices modulo a fixed integer larger than these consecutive-neighbor bounds. Each resulting class is an increasing matching. Symmetrization uses further fixed classes, and retaining a duplicate only in its first class preserves the simple graph. Evaluation, division, comparison, and duplicate tests use polynomially many bit operations and cells in $\log m$ per candidate. Enumerating sources and filtering by class emits each page in increasing order. Thus this use of the expander input requires neither a table depending on m nor a unit-cost operation on a growing word.

The depth interface is a machine-independent consequence of the previous paper. We state precisely the portion needed below.

Lemma 5.2 (Cheap whole-step depth for home-based walks)**.** Cited: Theorem B and the proof of Corollary B of [4]. Both the home-based cut theorem and the whole-step conversion used here are written proofs, not Lean-verified results.

Fix an oblivious native machine with k work tapes, whole-step value width at most β bits, and running time at most $c(L+1)$. Suppose every work-head walk is home-based: every greedy weakly monotone leg begins or ends within distance one of its fixed home cell. For every fixed $\varepsilon > 0$, and all sufficiently large admissible L, there is a whole-step genie satisfying $\beta|\Gamma| \le \varepsilon L$ whose full depth, in the sense of Definition 3.1, is at most a constant D. The constant may depend on the machine and ε, but not on L.

This is the cut theorem and long-edge conversion of [4, Theorem B and the proof of Corollary B], before any assumption of transposition correctness. A cut edge $t \to s$ is blocked by revealing s; a periodic set of additional revealed steps bounds the runs of state edges and pauses between long hops. The imported statement therefore controls all edges of the whole-step DAG, not just long hops. We will use it only for existence. Our conclusions about a specified depth-D genie quantify over every such genie.

Proposition 5.3 (A fixed native shear executor)**.** Not formalized; a complete proof is given below.
There is a fixed deterministic oblivious native machine M_0 which, at all sufficiently large even canonical dimensions, computes

$$\Phi(A, B) = (A, B + L_m A)$$

in $O(L)$ transitions, where A, B are the first and last m raw rows and L_m is the binary adjacency matrix of H_m. It uses read-only raw input, blank initial work tapes, and append-only raw output, and all work heads are home-based. There is also such a machine M_1 which appends all A rows only after completing the shear pages, followed by all B rows. Both machines have fixed alphabet, state set, and tape count.

Proof (not formalized). We give the execution details because later arguments use its visits, including visits at which the scanned symbols are semantically irrelevant. Two separate work tapes will hold banks of length $n = mW$. They begin blank; each receives a persistent origin flag, row-start flags during loading, and an end marker when its loading finishes. A unary row clock has length W and its own origin and end marker. All flags are tracks of a fixed finite alphabet. Both data heads remain at their origins throughout setup, so setup does not visit or initialize the full banks.

First count the raw input length with a little-endian binary counter whose head returns to its origin after each carry. The sum of carry lengths is $O(L)$, including return traversals. The raw input is then rewound. Canonical shape recovery uses polynomially many bit operations in $\log L$: try the possible entry widths, compute an integer square root, and check the defining equality. Invalid lengths, odd dimensions, and a fixed finite set of small cases use an identity fallback. Only the even canonical subfamily is used in this section.

All polylogarithmic-space arithmetic can be made physically home-based at polynomially larger polylogarithmic cost. To make this explicit, bootstrap a unary scratch ruler of length $S = (\lceil \log_2(L + 2) \rceil + c)^h$ by repeated complete sweeps. Here the fixed exponent h exceeds the space exponent of the routines to be simulated. A fixed number of tracks in this window represent virtual tapes, symbols, and virtual head flags. A complete outward-and-return scan reads the flagged symbols; further complete scans implement a virtual transition and relocate flags. Right moves can be flagged on an outward scan and left moves on a return scan, using old and new flag tracks. Each virtual transition costs $O(S)$, and every physical turn occurs at the origin or at the end of the ruler. The length counter itself is implemented by the direct carry routine, so this simulation does not multiply its linear cost by a polylogarithm.

Generate the J endpoint lists of Lemma 5.1 as monotone output streams. Copy an individual record to the origin buffer before parsing it. The long list heads make monotone traversals and full rewinds, rather than reversing locally for arithmetic. Their total length is $O(m \log m)$ and the compiler still costs $O(m \operatorname{polylog} L)$. Construct the unary row clock by reading the binary digits of W from most to least significant. At each digit, double the current unary string onto an alternate tape, append the digit, erase and rewind the old string, and rewind the new one. The geometric sum of the string lengths is $O(W)$. All these work heads are home-based.

Load the A bank and then the B bank in raw row order. The row clock identifies the first bit of each row, where the load writes the row-start flag; completion of a bank writes its end marker at n. Thus these flags require no preliminary full-bank sweep. An active bank head and the row-clock head advance one cell per bit. At a row boundary the bank head waits while the clock returns to its origin. The machine M_0 also appends each A bit at its load transition; M_1 omits these early appends. After loading, rewind both banks completely.

For each increasing matching page, read its next endpoint pair (i, j), skip whole rows forward until the two indicated rows are reached, and scan the pair in alignment. In a single transition at

offset s, read $A_i[s]$ and $B_j[s]$, write their XOR to $B_j[s]$, and advance both heads and the row clock. A skip likewise uses one transition per bit and preserves its payload. Clock rewinds and controller operations take place with the bank heads parked at row starts or at the bank end. Increasing endpoints make a single forward traversal sufficient. After the last matched pair, scan the unused suffixes to the end and rewind both banks in lockstep, one cell per transition. Empty pages use the same complete traversal convention. Each page costs $O(L)$.

All work cells scanned on every transition are rewritten, including cells under parked heads. Preserving a symbol is a write, and no part of the construction assumes otherwise. Induction over the pages shows that the B bank contains $B + L_m A$: every simple edge is executed exactly once, so duplicate XORs do not cancel a desired contribution. The A bank remains unchanged. Finally M_0 appends the B bank. For M_1, after the last full rewind, scan and append A and then B. This has the specified delayed output order.

The complete cost is

$$O(L + W + JL + m \operatorname{polylog} L + \operatorname{polylog} L) = O(L),$$

since $L = 4m^2 b$ and all logarithmic powers are fixed. Public parameters, flags, and endpoint records determine every move and append time; payload values determine only written or appended bits. The schedule is therefore oblivious. Bank and metadata heads make complete outward traversals and returns, possibly pausing along a traversal. Clock and compiled controller heads also make origin excursions. These are home-based physical walks. The number of pauses need not be bounded, and we make no assertion of a bounded number of affine pieces within a leg. □

5.2 Literal expander paths and robust separators

The native DAG has a state edge $t \to t - 1$ and, for each work tape, an edge from t to the previous visit to its scanned cell, if one exists. Its in- and out-degrees are at most $k + 1$ after parallel edges are merged. Raw input bits attach to their read steps; the read-only input tape does not contribute previous-visit edges. A genie-free path may start at a revealed vertex but has no subsequent revealed vertex. When proving survival below, we conservatively discard a path meeting the genie even at its first vertex.

Lemma 5.4 (A literal core with bounded congestion)**.** Not formalized; a complete proof is given below. *For either machine in Proposition 5.3, every edge A_iB_j of H_m and every lane $s \in [W]$ has a literal backward DAG path from the append of $B_j[s]$ to a read of raw input $A_i[s]$. The chosen paths have maximum vertex congestion at most $C_0 = 2\Delta$, including endpoints. For lanes $1 \le s < W$, their lengths are bounded by a constant depending only on J. No length bound is needed for lane zero.*

Proof (not formalized). Let t_{in} be the load of $A_i[s]$, t_{xor} the update for the chosen edge, and t_{out} the append of $B_j[s]$. List every visit to the physical $A_i[s]$ cell from t_{in} through t_{xor}; then list every visit to the $B_j[s]$ cell from t_{xor} through t_{out}. Consecutive visits in each list are actual previous-visit edges when read backward, and the lists share the XOR vertex. Their concatenation gives the claimed path to an actual raw read. It includes unchanged writes and controller waits.

For $s \neq 0$, no bank head parks at this cell. The A cell has at most $2J + 2$ relevant visits and the B cell at most $2J + 3$; a bound of $4J + 8$ on path length is more than sufficient. At lane zero, all parked visits are included and can make the path long. At any time a scanned A cell belongs to paths for at most Δ incident expander edges, and the same holds for the scanned B cell. Their union has size at most 2Δ. The extra delayed A scan in M_1 can add parked visits on the B origin, but does not change this congestion estimate. □

Thus E_Γ contains W vertex-disjoint copies of H_m with at most $C_0|\Gamma|$ missing core edges. There are $2mW = L$ core ports, a subset of the $2L$ incidence ports. The claim concerns containment in the full syntactic cone graph; unselected state and cell paths may supply further edges.

Proposition 5.5 (A robust separator estimate)**.** Not formalized; a complete proof is given below.
Let U be any deleted coordinate set. If a further port set Z makes every component of $E_\Gamma - U - Z$ have size at most $K \le m$, then

$$|Z| \;\ge\; \frac{\eta L - 2C_0|\Gamma|}{\Delta + \eta} - 2|U|.$$

This holds for every genie, without a depth hypothesis.

Proof (not formalized). Discard the at most $f = C_0|\Gamma|$ core edges whose paths meet Γ. At most $z \le 2|U| + |Z|$ core ports have been deleted. Intersect each remaining component with each lane core. These intersections have size at most m, so expansion applies to each. Summing their edge-boundary lower bounds counts at least $\eta(L - z)$ crossing edges with multiplicity. An edge joining two surviving intersections must be one of the f lost core edges, and is counted at most twice. Edges meeting the deleted core ports contribute at most Δz. Hence

$$\eta(L - z) \le 2f + \Delta z.$$

Rearrange and use $z \le 2|U| + |Z|$. Additional cone edges can merge components, but cannot invalidate this lower bound. □

5.3 The full syntactic cones retain a lane constraint

The preceding core alone says nothing about code rate. For that purpose we must control every cone edge, including those introduced by parked heads and state transitions. Fix either M_0 or M_1, let k be its number of work tapes, and write

$$B_D = \sum_{h=0}^{D} (k+1)^h.$$

Lemma 5.6 (A sparse interface for full cones)**.** Not formalized; a complete proof is given below.
There are fixed constants C_s, h_s and a set X of actual time vertices, depending only on the length-specific schedule, such that

$$|X| \le S_d := C_s\big[W + d(1 + \log_2(L+2))^{h_s}\big].$$

For every full-depth-D genie, every output outside the D-step forward neighborhood of X, and every non-row-start raw input in its cone, the two physical within-row bit offsets have cyclic distance at most D in $\mathbb{Z}/W\mathbb{Z}$.

Proof (not formalized). The entire length count, shape calculation, metadata generation, and clock construction form the setup. This can take $\Theta(L)$ transitions, so it cannot be discarded as a small set of times. Its distinct work cells, however, number only $O(W + d\,\mathrm{polylog}\,L)$. Mark the first post-setup state vertex and the first post-setup visit to every cell touched during setup. Any backward state or cell edge entering setup begins at one of these marked vertices.

Post-setup fast phases are a forward row load, skip, or XOR scan; a clock return; a simultaneous whole-bank rewind; and a final forward output scan. Mark all other controller, parser, arithmetic,

and metadata ticks. Their total is $O(d \operatorname{polylog} L)$. Mark also the first and last ticks of each fast phase, moving bank visits to offset zero modulo W, and moving clock visits to its endpoints. There are $O(d)$ such marks, with the fixed number of pages absorbed in the constant. A stationary head at offset zero is not marked at every tick.

Every unmarked post-setup vertex has a lane: the position of its moving clock modulo W, if the clock moves, and otherwise that of its moving bank. Simultaneously moving banks have the same lane. A state edge between unmarked vertices changes lane by at most one. For a cell edge, a stationary head has its immediately preceding time as predecessor once the first fast tick has been excluded. A moving bank head at a nonzero offset has no parked visits at that cell; its preceding post-setup moving visit has the same lane. The same is true of a moving interior clock cell, in a forward row scan or return. These cases cover all work tapes, including tapes whose fields the current operation ignores. Consequently every edge between good vertices changes lane by at most one cyclically.

The D-step forward neighborhood of X contains at most $B_D|X|$ time vertices, since forward degree is at most $k+1$. An output outside it has no marked vertex in a depth-D free cone. Its cone also cannot enter setup, by the first-touch marks. During loading, a non-row-start raw bit is read only at a load vertex with that same lane. During row-boundary pauses the raw head holds the next row-start bit or is outside the payload; after loading it is outside the payload. Excluding the d raw row starts therefore accounts for every remaining input attachment. Summing lane changes along a free path proves the result. The output itself is included in the forward-neighborhood exclusion, which covers the convention allowing a free path to start at a revealed vertex. □

The following elementary code argument will be useful for both examples.

Lemma 5.7 (An order of port colors)**.** Lean: `color_order_half_bound` [`ColorOrder.lean`]

Let a paired incidence graph on N coordinates admit colors $\alpha(v)$ at L_v and $\zeta(v)$ at R_v such that every edge is monochromatic and $\alpha(v) \neq \zeta(v)$ for every v. Every storage code over an alphabet of cardinality a has size at most $a^{\lfloor N/2 \rfloor}$.

Outline of the formal proof. Order the finitely many used colors. Record the coordinates v for which $\alpha(v)$ precedes $\zeta(v)$. Process left-color classes in increasing order. If a coordinate was not recorded, then its right color precedes its left color. Every coordinate in its recovery neighborhood has that earlier left color, by monochromaticity, and so has already been recovered. This proves that projection onto the recorded coordinates is injective on the code; no extension of an arbitrary partial assignment is needed. Compare the color order with its reverse. Each coordinate is recorded in exactly one of them, so one record has at most $\lfloor N/2 \rfloor$ coordinates, which proves the claim. □

Theorem 5.8 (Half rate on every induced full-cone relation)**.** Not formalized; a complete proof is given below.

Fix $D \geq 0$. For either native machine in Proposition 5.3, for every whole-step genie of full depth at most D, every $Q \subseteq V$, and every storage code $\mathcal{C} \subseteq \mathcal{A}^Q$ for the induced actual cone relation, where $a = |\mathcal{A}| \geq 2$ and $N = |Q|$, there is $u \leq \min(N, e_D(d))$ such that

$$|\mathcal{C}| \leq a^{\lfloor (N+u)/2 \rfloor}.$$

One permissible error bound, uniform in the genie, alphabet, and Q, is

$$g = \lceil \sqrt{d} \rceil, \qquad r = \lceil d/g \rceil,$$
$$e_D(d) = B_D S_d + d + 2d(2D+1)r + bdg = o_D(L).$$

There is no bound on the cost of the genie in this theorem.

Proof (not formalized). Partition the column indices into r consecutive buckets of at most g columns each, and view their bit lanes as consecutive arcs of $\mathbb{Z}/W\mathbb{Z}$. Guard all lanes within cyclic distance D of a bucket boundary. There are at most $(2D+1)r$ guarded bit lanes. Delete from V the following coordinate sets: demand labels of the bad physical outputs in Lemma 5.6; the d raw row starts; coordinates whose input or transpose-output lane is guarded; and coordinates (i,j,s) whose row and column belong to the same bucket. The first set costs at most $B_D S_d$, the third at most $2d(2D+1)r$, and the fourth at most bdg. The total deletion is at most $e_D(d)$.

On the surviving ports give $L_{(i,j,s)}$ the bucket of j, and give $R_{(i,j,s)}$ the bucket of i. A surviving incidence edge has equal colors by Lemma 5.6: a path of lane length at most D cannot cross a bucket boundary while both endpoint lanes avoid its guard. The own-port colors differ by the fourth deletion. The graph therefore satisfies Lemma 5.7.

Intersect the deleted set with Q, obtaining u coordinates. Condition their values in the original code. Each nonempty fibre is a storage code on the induced surviving relation: the removed recovery inputs have fixed values in that fibre. Lemma 5.7 bounds its size by $a^{\lfloor (N-u)/2 \rfloor}$. There are at most a^u fibres, giving $a^{u+\lfloor (N-u)/2 \rfloor} = a^{\lfloor (N+u)/2 \rfloor}$. For fixed D, each displayed error divided by $L = d^2 b$ tends to zero. All exclusions were chosen from the schedule and the full relation; the argument is simultaneous for all alphabets, recovery functions, and codes. □

Corollary 5.9 (Nonhyperfinite bounded-degree cones with half rate)**.** Not formalized; a complete proof is given below.

For one fixed native home-based linear-time machine there are constants $\varepsilon, \alpha > 0$ and fixed integers D, M with the following property. At every sufficiently large even canonical dimension the class of genies satisfying $\beta|\Gamma| \le \varepsilon L$ and full depth at most D is nonempty. For every such genie, one can delete coordinates to obtain an induced paired graph on $N = \Theta(L)$ coordinates, of maximum degree at most M, with no own-port edge, such that:

1. *for every fixed K and all sufficiently large d, every port set whose deletion leaves components of size at most K has size at least $\alpha L/2$;*
2. *every storage code on every further induced coordinate set has logarithmic cardinality at most half its coordinate count plus $o_D(L)$, uniformly over all finite alphabets.*

Proof (not formalized). Choose $\gamma > 0$ with $\gamma \le \eta/(8C_0)$, and then $0 < \varepsilon \le \min(1, \beta\gamma)$. Set $\alpha = \eta/[2(\Delta+\eta)]$. Proposition 5.5 gives a separator lower bound at least αL before coordinate trimming for every genie of the stated price. Lemma 5.2 supplies a fixed D and makes the permitted genie class nonempty.

Each right incidence degree is at most B_D, so there are at most $B_D L$ edges. Choose a constant degree threshold $M_0 \ge 8B_D/\alpha$. Delete every coordinate whose left degree exceeds M_0; their number is at most $\alpha L/8$. Also delete coordinates with an own-port edge. The one-stage transpose-matching estimate of Corollary 3.7 bounds these by $O(L/\log d)$, uniformly over the genies in question; eventually this is at most $\alpha L/8$. This use of the matching estimate does not require transpose correctness or an initial left-degree bound. The remaining graph has maximum degree at most $M = \max(M_0, B_D)$ and $N = \Theta(L)$. The separator loss is at most twice the number of deleted coordinates, giving $\alpha L/2$. For any fixed K, eventually $K \le m$, so Proposition 5.5 applies. The code conclusion is Theorem 5.8, whose quantifiers already include every induced coordinate set. □

Remark 5.10. The conclusion rules out a uniform small-component decomposition for this family at arbitrarily small deletion cost. It does not exhibit a high-rate code. In fact its simultaneous half-rate bound shows why a directed coding theorem can remain possible after a separator approach fails. Neither the lane lemma nor its coloring is claimed for arbitrary native schedules. The next

construction changes the paid physical layout and destroys almost-everywhere separation of paired ports by component colors.

6 Connected demand pairs and a depth-independent coding deficit

The coloring in the preceding section separates the two ports of almost every coordinate. We now give a paid native modification for which no such separation by connected components is possible after a sublinear coordinate deletion. Nevertheless its full cone relations have a positive all-code deficit whose coefficient is independent of the depth. The proof uses a finite row-phase description and an independent rectangle, and does not recover a half-rate bound.

We retain $d = 2m$, $b = \max(1, \lceil \log_2 d \rceil)$, $W = db$, $L = d^2 b$, the graphs H_m, and the physical transpose-demand labels of Section 5. Indices of B rows in this section are global: $m \le j < d$. Define a row-dependent rotation by

$$R : (i, c, s) \longmapsto (i, c + i \pmod d, s).$$

The new machine computes the invertible map $R^{-1}\Phi R$. Throughout this section, k and β denote the work-tape count and a whole-step value-width bound for this rotated machine M_R, including its additional buffers and controller tapes. They need not equal the corresponding bounds for the unwrapped machines. We use $B_D = \sum_{h=0}^{D} (k+1)^h$ with these enlarged resources.

6.1 A robust count of connected pairs

Lemma 6.1 (Prescribed pairs in damaged expanders)**.** `Lean`: `DamagedExpanders.damaged_expander_copies` [`DamagedExpanders.lean`]

Let $H_1, \dots, H_W$ be disjoint graphs on $2m$ vertices, each of maximum degree at most Δ and with outside vertex expansion at least $\eta|S|$ for every $|S| \le m$, where $\eta > 0$. Prescribe n_* pairs of distinct vertices, with all $2n_*$ endpoints distinct and each pair contained in one graph. After deletion of f edges and z vertices, at least

$$n_* - \frac{f}{\eta} - \left(1 + \frac{\Delta}{\eta}\right) z$$

prescribed pairs survive with their endpoints in the same connected component. The bound holds for every finite $m \ge 1$ and arbitrary distributions of the deletions among the graphs.

Outline of the formal proof. Vertex expansion implies an edge-boundary bound $|\partial_E S| \ge \eta|S|$. Consider one graph, and write f_0, z_0 for its deletion counts. If the surviving graph has a component G of more than m vertices, let S be the union of its other surviving components. Then $|S| < m$. Every original edge from S to G was deleted, and at most Δz_0 edges go from S to deleted vertices. Thus

$$\eta|S| \le f_0 + \Delta z_0.$$

Every prescribed pair wholly inside G is connected. All other pairs can be charged to distinct endpoints in S or the deleted set, since the prescribed endpoints are disjoint. Their number is at most $f_0/\eta + (1 + \Delta/\eta) z_0$.

If every surviving component has size at most m, sum the edge-boundary inequality over those components. Edges between surviving components were deleted and are counted at most twice; edges incident to a deleted vertex contribute at most Δz_0. Hence

$$\eta(2m - z_0) \le 2f_0 + \Delta z_0, \qquad m \le \frac{f_0}{\eta} + \frac{\Delta + \eta}{2\eta} z_0.$$

There were at most m prescribed pairs in this graph. Declaring all of them lost therefore satisfies the same asserted bound. This includes an empty surviving graph. Summing over the copies proves the lemma. □

6.2 A physical rotation wrapper

Proposition 6.2 (The rotated native shear)**.** Not formalized; a complete proof is given below.

There is one fixed deterministic oblivious native machine M_R which computes $R^{-1}\Phi R$ in $O(L)$ transitions at all sufficiently large even canonical dimensions. All work heads are home-based. For each edge A_iB_j of H_m and each internal lane $(g, s) \in [d] \times [b]$ there is a literal backward path from the final physical append

$$(j, g - j \pmod d, s) \quad \text{to a raw read at} \quad (i, g - i \pmod d, s).$$

These paths have endpoint-inclusive whole-step vertex congestion at most $C_1 = 4\Delta$. Their lengths need not be bounded.

Proof (not formalized). Use the two banks and the paid setup and page executor of Proposition 5.3. Add two reusable physical row-buffer tapes, denoted $P_{\mathrm{buf}}, S_{\mathrm{buf}}$, each with an origin and a movable end marker. Their symbols have fixed bit, origin, and end tracks.

We first specify a paid split-and-concatenate primitive. Given a public cut $a \in [0, W]$, clear the previous cut marker by a complete outward and return sweep of the row clock. Initialize an origin-based binary countdown to a, move the clock one cell after each decrement, mark the cut when the countdown becomes zero, and rewind the clock. The countdown does not perform a full zero test at each decrement: mark its highest one, update that mark when a borrow removes it, and remember zero when no one remains. The total digit traversal in a countdown of a is $O(a + \log(a + 1))$, including returns to the low digit. Initial parameter arithmetic uses the complete-sweep compiler of the preceding section. The cut therefore costs $O(W + \operatorname{polylog} L)$, and all these head motions are home-based.

To split a sequential W-bit source row, scan its bits with the row clock. In one transition per bit copy into P_{buf} before the cut and into S_{buf} at or after it, advancing only the source, active buffer, and clock. A fixed mode changes at the cut flag. Mark the two buffer ends and rewind the buffers and clock completely. An empty buffer has its end marker at zero. On reuse, writing a bit clears any former end flag at that cell; if a shorter string is stored, its new first end marker stops subsequent reading. Cells beyond that end are neither read nor erased. To concatenate in rotated order, scan S_{buf} and then P_{buf}, copying to the sequential destination or directly appending to the real output. Each bit transfer is one native transition, and the buffers are fully rewound afterward. There is no long-lived payload register and at most one output bit per transition. The entire primitive costs $O(W + \operatorname{polylog} L)$.

For input row i, split at $W - ib$ and concatenate S_{buf} followed by P_{buf} into its bank row. This is a right rotation by ib bits, so raw column c reaches internal column $c + i \pmod d$ without changing its within-entry bit offset. Process all rows in order, loading A and then B, with no output appends. Each bank head moves monotonically through its bank and pauses during the buffer operations. Write its row-start flags on the first concatenated bit of each row and its end marker on completing the bank; no full-bank setup sweep is used. After loading, rewind both banks completely. Execute the J shear pages exactly as in Proposition 5.3, including suffix scans and complete synchronized rewinds.

Finally process bank rows $i = 0, \ldots, d - 1$ in order. Split the bank row at ib, and append S_{buf} followed by P_{buf}. This is a left rotation by ib bits and implements R^{-1}. A source bank row is copied

into buffers before its output is emitted. In particular, even the unchanged A output is delayed until after every core page. We do not give its syntactic cone an unjustified restriction to A inputs. The resulting B output row is, explicitly,

$$B_j[c,s] + \sum_{i:A_iB_j\in E(H_m)} A_i[c+j-i \pmod d, s],$$

with addition modulo two.

There are d input and d output primitives. All parameter calculations, cut marking, countdown borrows and returns, data copies, buffer rewinds, bank rewinds, and appends are included in

$$O(L + dW + d\,\mathrm{polylog}\,L + m\,\mathrm{polylog}\,L + \mathrm{polylog}\,L) = O(L).$$

The fixed page family, compiler, counters, buffers, and clocks use a fixed number of tapes and fixed control. End flags and public row indices determine all movements, independently of the payload. Banks traverse monotonically from origin to end, with pauses, and return completely; buffers and clocks make origin-to-end excursions; counters return to their own origins. Thus the entire physical trace is home-based. During concatenation a bank may pause inside a row, so the row-start parking property of the unwrapped executor is not claimed here.

It remains to certify the actual paths. Fix one edge and internal lane. Its raw source bit is first written to an input-buffer cell, then copied from that same cell to $A_i[gb+s]$. The core XOR copies its contribution to $B_j[gb+s]$; in the final wrapper that bank cell is copied to an output-buffer cell and then read for the actual append. Follow all consecutive visits in these four residence intervals: input buffer to bank copy, A bank to XOR, B bank to output-buffer copy, and output buffer to append. Include every parked rewrite. Consecutive visits are previous-visit edges, and successive intervals meet at the same copy or XOR transition. Reversing their concatenation gives a literal DAG path with the displayed physical endpoints.

At any time each active bank carrier participates in at most Δ chosen paths. The same holds for a live buffer carrier, because it belongs to one source or one final target bit. Reuse of a buffer cell does not accumulate congestion: its residence intervals for different rows are disjoint, and each path has left the cell before it is reused. The input and output phases are disjoint, and buffering an A output carries none of the selected paths. A step scans one cell of each of the four payload tapes, so the union bound gives congestion 4Δ, including steps at which a path changes carrier and all parked endpoints. □

Theorem 6.3 (Many robustly connected demand pairs)**.** Not formalized; a complete proof is given below.

For M_R, every genie Γ and every deleted coordinate set U leave at least

$$\frac{L}{4} - \frac{C_1}{\eta}|\Gamma| - 2\left(1+\frac{\Delta}{\eta}\right)|U| \tag{6.1}$$

surviving coordinates whose two ports are connected in the induced full cone graph. There are constants $\varepsilon > 0$ and fixed integers D, M for which the permitted cheap depth-D genies are nonempty and, for every such genie, the usual degree and own-port-edge coordinate trims yield a bounded-degree graph on $N = \Theta(L)$ coordinates with at least $L/8$ connected demand pairs. After any additional $o(L)$ coordinate deletion at least $L/16$ such pairs remain for all sufficiently large d.

Proof (not formalized). For each internal lane (g,s), the literal paths give a copy of H_m with core ports

$$A_i \longleftrightarrow L_{(i,g-i,s)}, \qquad B_j \longleftrightarrow R_{(g-j,j,s)},$$

where columns are reduced modulo d. These W cores have L distinct ports. For a coordinate $p = (i, c, s)$ with $i < m$ and $c \geq m$, its two ports lie in the same core lane $g = i + c \pmod d$. Prescribe these $m^2 b = L/4$ disjoint pairs. Their distribution among lanes need not be uniform.

Discard every selected core path meeting Γ, including endpoints. At most $C_1|\Gamma|$ edges are discarded. Every remaining path witnesses an actual full-cone incidence. Deleting U removes at most $2|U|$ core ports. Lemma 6.1 proves (6.1), and extra full-cone edges can only join components.

Put $A_* = C_1/\eta$ and $B_* = 2(1 + \Delta/\eta)$. Choose $0 < \varepsilon \leq \min(1, \beta/(16A_*))$. For every genie of price $\beta|\Gamma| \leq \varepsilon L$, the first loss in (6.1) is at most $L/16$. Lemma 5.2 supplies a fixed full depth D and nonemptiness. Fix this D and let $B_D = \sum_{h=0}^{D}(k+1)^h$. Choose $M_0 \geq 32B_*B_D$. There are at most $B_D L$ incidence edges, so deleting coordinates with left degree above M_0 costs at most $L/(32B_*)$. The self-demand estimate of Corollary 3.7 bounds the own-port-edge deletion by $O(L/\log d)$, eventually also at most $L/(32B_*)$. The remaining maximum degree is at most $M = \max(M_0, B_D)$ and its coordinate count is at least $L - L/(16B_*)$. Formula (6.1) now leaves at least $L/4 - L/16 - L/32 - L/32 = L/8$ pairs. An additional set U' costs at most $B_*|U'|$ in that formula, proving the last assertion.

The choices have the order machine, ε, D, M_0, and finally sufficiently large d. None of D, M_0 grows with dimension. All conclusions hold for every genie satisfying the fixed price and depth bounds, rather than for a selected favorable genie. □

An edge-monochromatic port coloring is constant on each connected component. Thus Theorem 6.3 precludes separating all surviving own-port pairs by such a coloring after $o(L)$ further deletion, even after paying the fixed degree and self-demand trims. This does not contradict the directed incidence bridge: for every fixed radius, that bridge bounds the number of short paired paths by $O(L/\log d)$. Almost all of the connected pairs above consequently have distance exceeding any prescribed fixed radius. Connectivity and bounded-distance recovery are different assertions.

6.3 Every full cone has a fixed row-phase description

We next analyze all syntactic dependencies of the same machine, including the auxiliary tapes. We do not infer an incidence upper bound from the selected expander paths or from the support of the shear matrix.

A *row word* with ℓ maps has the form

$$x_0 \xrightarrow{\text{band}} y_0 \xrightarrow{f_1} x_1 \xrightarrow{\text{band}} y_1 \xrightarrow{f_2} \cdots \xrightarrow{f_\ell} x_\ell \xrightarrow{\text{band}} y_\ell.$$

There are exactly $\ell+1$ bands, including the endpoint bands. Each map is a strictly monotone partial injection between integer intervals; each band permits an arbitrary subset of the pairs $|x - y| \leq r$ in its interval. For $\ell = 0$ the word is one band. Different maps may have different intermediate domains. A word records every attained endpoint, and need not be single-valued.

Lemma 6.4 (A fixed number of row words)**.** Not formalized; a complete proof is given below.

For the complete trace of M_R there are fixed integers $K, \ell_{\max}$ and fixed constants C, h with the following property. For every fixed $D \geq 0$ and every full-depth-D whole-step genie, there is a coordinate exception set Z with

$$|Z| \leq z_D(d) := B_D C\big[W + d(1 + \log_2(L+2))^h\big] + d = o_D(L)$$

such that every actual cone incidence $L_p R_q$ with $p, q \notin Z$ has its physical output row $\mathrm{outrow}(q)$ *and raw input row* $\mathrm{rawrow}(p)$ *related by one of at most K row words. Each word has at most $\ell_{\max}$*

strictly increasing partial injections and bands of radius $r = D + 2$. Its direction is from physical output row to raw input row. One may take

$$M_{\mathrm{ph}} = 2J + 3, \qquad K = 3^{M_{\mathrm{ph}}} + 1, \qquad \ell_{\max} = 2(M_{\mathrm{ph}} - 1).$$

The maps can depend on the dimension's actual scan events; their number and the word lengths do not depend on d, D, or the genie.

Proof (not formalized). Call everything through parameter, endpoint-list, and clock initialization setup. Thereafter the successive macro phases are

$$I,\ R_0,\ F_1,\ R_1, \ldots,\ F_J,\ R_J,\ O.$$

Here I is the complete input wrapper, O the complete output wrapper, R_0 the post-load synchronized rewind, F_a the forward part of page a, and R_a its synchronized rewind. There are M_{ph} phases, each occurring once. A backward edge crossing phases strictly decreases the phase index.

In a wrapper phase, row episode i contains its parameter calculation, cut marking, countdown, split, concatenation, tests, and returns. Episodes occur in increasing global row order and do not interleave. In a forward page, a scan event consists of a whole W-bit A skip, B skip, or matched A/B scan and its following clock return. Number events in chronological order, ignoring intervening controller ticks. There are at most $2m = d$ events in one page. Each row of either bank is active in exactly one event and its active events are in increasing order. Let

$$\tau_{a,A}, \tau_{a,B} : [m] \longrightarrow [E_a]$$

be the corresponding strictly increasing injections. Their values agree at a matched scan. One common event index is used even when one bank is parked; the parked cursor is not an independently selectable row.

We define a sparse set X of marked real time vertices. First mark the first post-setup state vertex and the first post-setup visit to each work cell touched in setup. The setup inventory, not its time, is $O(W + d \operatorname{polylog} L)$. These marks block every backward edge entering setup. Next mark the first and last ticks of every macro phase and wrapper row episode, all synchronized-rewind turn and end-test ticks, all slow controller, parser, list, and arithmetic ticks in core pages, and the first and last ticks of each core row scan and clock return. The core slow work totals $O(d \operatorname{polylog} L)$. Wrapper countdowns are not included in this wholesale marking: their time can be linear.

On each bank mark moving visits to row-boundary cells, and the first tick of each maximal constant-position run of at least two ticks. There are $O(d)$ such runs per fixed phase: a bank pauses at the constant number of operation boundaries in each wrapper row, and during core row or controller operations. A long countdown gives one bank wait, not a new wait at each decrement. This rule also covers an interior bank pause between the two concatenation segments.

For each macro phase, mark its first visit to each cell of the row clock and the two row buffers. They occupy only $[0, W]$, giving $O(M_{\mathrm{ph}} W)$ marks. Finally, for each wrapper row episode mark its first visit to each cell of every other controller or work tape. Those tapes use only polylogarithmically many distinct cells in that episode: metadata heads park at their origins during wrappers, and the scratch ruler, arithmetic, and countdown use their bounded polylogarithmic windows. This costs $O(d \operatorname{polylog} L)$. We have not marked first visits to every bank cell in every phase, nor every clock cell in every row. Either would incorrectly charge a linear number of vertices. The stated rules give

$$|X| \le C[W + d(1 + \log_2(L + 2))^h].$$

Assign each unmarked post-setup vertex a chart: its global row-episode index in I or O, its event index in F_a, and the common local bank row $\lfloor \text{position}/W \rfloor$ in R_a. At good rewind vertices both banks move in lockstep through interior cells. We check that every same-phase edge between good vertices changes chart by at most one.

For state edges this follows from chronological row or event order. For a wrapper controller tape, an edge into an earlier episode starts at the first visit to that cell in the current episode and was marked. All remaining such edges stay in their row, including every countdown borrow and return. The row clock receives a complete outward-and-return sweep while clearing the cut in every wrapper row. A clock predecessor therefore lies in the same or preceding row episode. Its first macro-phase touch is already marked.

For buffers, every used prefix, including its end, is visited in the row episode. During I, the P_{buf} lengths $W - ib$ decrease and the S_{buf} lengths ib increase; during O these monotonicities are reversed. If a cell has an earlier visit within a wrapper phase, this monotonicity ensures that the immediately preceding episode also visited it, unless it has already been visited in the current episode. Position zero is visited even for an empty buffer. The first touch in each macro phase was marked, so stale earlier-phase visits cause no exception.

For a wrapper bank, any stationary visit after the first has predecessor $t - 1$, while the first tick of a nontrivial parking run was marked. An isolated moving interior visit belongs to its physical row episode. The bank progresses monotonically through that wrapper, so an earlier same-phase visit to that cell is in the same row. Moving row-boundary visits were marked. These facts cover the possible interior pauses.

At good core forward vertices, controller tapes and buffers are stationary; after the marked first fast tick their predecessors are $t-1$. The same holds for a parked bank after its marked first parking tick. Each active interior bank cell is visited once in the forward page, so any other predecessor leaves the phase. The moving clock visits each interior cell on the scan and return of every event, giving a predecessor in the same or preceding event. In a synchronized rewind, each bank cell is visited once, all other heads are stationary, and state edges change local row by at most one. Turn and end-test vertices were marked. This completes the same-phase edge audit.

Now consider an edge with good endpoints in different phases. State crossings are blocked by phase marks, clock and buffer crossings by their first macro-phase touches, and controller crossings by per-row first touches in wrappers or by stationary/slow-tick rules in the core. Setup crossings have also been blocked. Only bank cell edges remain. The current endpoint cannot be the first tick of a parking run or a moving row boundary. It therefore has the chart of its actual active interior bank row. At the earlier endpoint, any wrapper pause at that interior cell belongs to the same physical row; a page or rewind has its unique active visit. Thus both endpoints refer to one and the same local bank row $u \in [m]$.

For bank $H \in \{A, B\}$ define the strict injections

$$\begin{aligned} &\iota_{I,A}(u) = \iota_{O,A}(u) = u, \qquad \iota_{I,B}(u) = \iota_{O,B}(u) = m + u, \\ &\iota_{R_a,A}(u) = \iota_{R_a,B}(u) = u, \quad \iota_{F_a,A}(u) = \tau_{a,A}(u), \qquad \iota_{F_a,B}(u) = \tau_{a,B}(u). \end{aligned}$$

A backward bank edge from phase F to an earlier phase F' acts exactly by $\iota_{F',H} \circ \iota_{F,H}^{-1}$ on its chart, with the inverse interpreted on its image. This factorization uses a common physical cell and does not assume that an increasing partial map preserves distances.

Every append occurs in O with chart equal to its physical output row. After input loading the raw head reads blank. During I it reads a bit of the current row while splitting, or waits at the first bit of the next row during concatenation and controller work. Exclude the d raw row-start coordinates. Every other raw attachment at a good vertex then has raw row equal to its chart.

Exclude also the demand labels of outputs in the D-step forward neighborhood of X in the full DAG. There are at most $B_D|X|$ of them. A free cone of any remaining output can meet neither X nor setup, because its paths have length at most D. The physical output exclusions are converted through Π to coordinate labels before being united with the raw row-start exclusions. This gives the set Z in the statement.

Break a surviving output-to-raw path at its cross-phase edges. Each same-phase segment has chart displacement at most D. Each cross-phase edge contributes the two strict partial injections just displayed. Since phase indices strictly decrease, there are at most $M_{\mathrm{ph}} - 1$ crossings. Enlarging every local displacement and every zero gap between these injections to a band of radius $D + 2$ gives the asserted words. There are at most $2 \cdot 3^{M_{\mathrm{ph}}-2}$ phase-and-bank choices for paths from O to I; the larger value $K = 3^{M_{\mathrm{ph}}} + 1$ is convenient and also covers empty or padded choices. The bounds on $K, \ell_{\max}$ are independent of depth. Finally $W/L = 1/d$ and $d \operatorname{polylog} L/L \to 0$, proving the fixed-$D$ error estimate. □

6.4 Threshold pullback through partial monotone words

Partial increasing injections need not preserve distances: the map sending $0, 100$ to $0, 1$ is a counterexample. For the row words it is sufficient to pull back thresholds, which does remain valid in the presence of arbitrary domain gaps.

Lemma 6.5 (Finite threshold pullback)**.** Lean:
`ThresholdPullback.finite_threshold_pullback` [`ThresholdBuckets.lean`]

Let K row words connect the starting and ending intervals $[d]$, each using at most $\ell_{\max}$ strictly monotone partial injections and bands of integer radius $r \geq 0$. Maps may be increasing or decreasing. Partition each endpoint interval into B nonempty consecutive buckets, where $1 \leq B \leq d$; the partitions may differ. There is a set of starting rows T with

$$|T| \leq \min\{d,\ K(B-1)[(2r+1)^{\ell_{\max}+1} - 1]\}$$

such that, for each word, all attained endings of any remaining starting row lie in one ending bucket. That bucket index is monotone in the starting row, with sign equal to the product of the map signs. In particular the union of all retained word relations occupies at most $K(2B-1)$ starting/ending bucket pairs.

Outline of the formal proof. A threshold or its complement pulls back through a strictly monotone partial injection to a cut on its domain, with polarity multiplied by the sign of the map. Extend that cut across domain gaps. Empty domains and constant predicates are included.

At a radius-r band, a cut can change only on at most $2r$ starting rows. Each exceptional row at the next layer has at most $2r + 1$ predecessors through a band and at most one through a partial injection. Backward induction through a word with ℓ maps therefore gives

$$E_0 = 2r, \qquad E_{j+1} = (2r+1)E_j + 2r, \qquad E_\ell = (2r+1)^{\ell+1} - 1.$$

Outside the charged set the pulled-back comparison is exact for every attained endpoint, simultaneously. The formal relation permits arbitrary subrelations of the full-band word, so omitted paths cannot invalidate this conclusion.

Take the union over K words and the $B-1$ ending-bucket boundaries. The boundary comparisons force one ending bucket for every retained starting row that attains an endpoint, and that bucket is monotone with the product sign. For a negative word reverse the ending-bucket order. Each retained occupied pair then lies in a coordinatewise chain in $[B] \times [B]$. The sum of its two indices

increases strictly between distinct comparable pairs and ranges over $2B-1$ values. Thus one word occupies at most $2B-1$ pairs, and their union at most $K(2B-1)$. □

In Lemma 6.4 the starting row is the physical output row. Consequently T in Lemma 6.5 is a set of output rows. Under transpose demand identification, deleting that row set means deleting coordinates (i,j,s) with $j \in T$, at cost $db|T|$ before overlaps. Swapping this with the raw row index would be an incorrect identification of the demand.

6.5 An independent rectangle and the all-code gap

Lemma 6.6 (A finite independent rectangle)**.** Lean: `IndependentRectangle.independent_rectangle`, `IndependentRectangle.independent_rectangle_consecutive` [`IndependentRectangle.lean`]

Let any directed recovery relation on $V = [d]^2 \times [b]$ have the following endpoint interface. There are raw-row exceptions S, physical output-row exceptions T, and coordinate exceptions Z. Partition $[d]$ into B contiguous near-equal buckets and suppose every incidence $L_p R_q$ with $p, q \notin Z$, $\mathrm{rawrow}(p) \notin S$, *and* $\mathrm{outrow}(q) \notin T$ *belongs to one of at most $K(2B-1)$ raw-row/output-row bucket pairs. Let $K \geq 1$, $B = 4K$, and $d \geq B$. Then the relation has an independent coordinate set I with*

$$|I| \geq L_0 := \max\left\{0, b(2K+1)\left\lfloor \frac{d}{4K} \right\rfloor^2 - bd(|S|+|T|) - |Z|\right\}.$$

Independence means that no recovery neighborhood of a coordinate of I contains a coordinate of I, including itself.

Outline of the formal proof. In the bipartite graph of occupied bucket pairs, some output bucket has degree at most

$$\left\lfloor \frac{K(2B-1)}{B} \right\rfloor = 2K-1.$$

Let C be its row set, and let R be the union of its nonneighboring raw buckets. At least $2K+1$ raw buckets are in R. Each bucket has at least $\lfloor d/(4K) \rfloor$ rows. Define

$$I = ((R \setminus S) \times (C \setminus T) \times [b]) \setminus Z.$$

Before exceptions its size is at least the leading term in L_0. The raw-row and output-row exclusions cost at most $bd|S|$ and $bd|T|$, and coordinate exclusions at most $|Z|$. This bound permits all exceptions to concentrate in the selected rectangle.

If $p, q \in I$ and p were in the recovery neighborhood of q, the interface would put $\mathrm{rawrow}(p) \in R$ and $\mathrm{outrow}(q) \in C$ in an occupied pair, contrary to the choice of R. Thus I is independent in the original relation; no individual incidence edge has been removed. □

Lemma 6.7 (Independent coordinates bound every induced code)**.** Lean: `induced_storageCode_card_le_of_independent`, `induced_storageCode_entropy_of_independent` [`StorageCode.lean`]

If I is independent in a directed recovery relation on V, then for every $Q \subseteq V$ and every storage code $\mathcal{C} \subseteq \mathcal{A}^Q$ for the induced relation,

$$|\mathcal{C}| \leq a^{|Q|-|I\cap Q|}, \qquad a = |\mathcal{A}|.$$

For any random word supported on the code, with any probability law,

$$H(X_Q) = H(X_{Q\setminus I}) \leq (|Q| - |I \cap Q|)\log_2 a.$$

Outline of the formal proof. Projection onto $Q \setminus I$ is injective on the code. Indeed, two codewords agreeing there agree on the entire recovery neighborhood of each coordinate in $I \cap Q$, since that neighborhood contains no coordinate of I. The storage-code property forces agreement on every remaining coordinate. This proves the cardinality and entropy assertions directly, without an extension assumption or a sequential recovery order. □

Theorem 6.8 (A positive deficit independent of depth)**.** Lean (partial): `row_word_cover_independent`, `row_word_cover_code_log_bound`, `row_word_cover_asymptotic_code_gap` [RowWordDeficit.lean]
Not formalized: The native row-word extraction and its uniform sublinear exceptional-count estimate.

There is a constant $\rho > 0$, depending only on the fixed machine M_R, with the following property. For every fixed D, every whole-step genie of full depth at most D, every induced coordinate set $Q \subseteq V$, and every nonempty storage code $\mathcal{C} \subseteq \mathcal{A}^Q$ for the actual full-cone relation, put $N = |Q|$, $a = |\mathcal{A}| \geq 2$. Then

$$\log_a |\mathcal{C}| \leq N - \max\{0, L_D(d) - |V \setminus Q|\},$$

where, for all $d \geq 4K$, one may take

$$t_D = K(4K-1)\big[(2D+5)^{\ell_{\max}+1} - 1\big],$$
$$L_D(d) = \max\left\{0, b(2K+1)\left\lfloor \frac{d}{4K} \right\rfloor^2 - bd\min(d, t_D) - z_D(d)\right\},$$

with $K, \ell_{\max}, z_D$ from Lemma 6.4. In particular one can choose $\rho = 1/(8K)$, independently of D, and obtain

$$\log_a |\mathcal{C}| \leq N - \rho L + |V \setminus Q| + o_D(L),$$

along with the trivial bound $\log_a |\mathcal{C}| \leq N$. No price bound on the genie, linearity assumption on the code, or degree bound on the relation is required.

Proof (not formalized). Apply Lemma 6.4, and partition the two row orders into $B = 4K$ near-equal buckets. Lemma 6.5 with $r = D + 2$ supplies at most $\min(d, t_D)$ exceptional starting rows, which are physical output rows. Transposing the occupied bucket pairs gives the raw-row/output-row interface of Lemma 6.6; the number of occupied pairs remains at most $K(2B-1)$. There are no extra raw-row exceptions, and the native coordinate exceptions number at most $z_D(d)$. Thus one independent set I, chosen from the relation alone, has size at least $L_D(d)$.

Since $|I \cap Q| \geq \max\{0, L_D(d) - |V \setminus Q|\}$, Lemma 6.7 proves the exact finite bound. The same I works for every Q, alphabet, and code on this relation. For the asymptotic assertion use $\lfloor x \rfloor^2 \geq x^2 - 2x$ for $x \geq 0$. The leading term satisfies

$$b(2K+1)\left\lfloor \frac{d}{4K} \right\rfloor^2 \geq \frac{2K+1}{16K^2} L - \frac{2K+1}{2K} db.$$

Its coefficient exceeds $1/(8K)$. For fixed D, the output-row loss is $O_D(db) = o_D(L)$ and $z_D(d) = o_D(L)$. This proves the asserted choice of ρ. The eventual dimension threshold can depend on D, while ρ cannot and does not. The empty code satisfies the cardinality bound separately. □

Remark 6.9 (What the gap establishes)**.** The same native trace simultaneously has the robust connected demand pairs of Theorem 6.3 and the all-code deficit of Theorem 6.8. Its independent rectangle sacrifices a fixed positive fraction of coordinates. It forbids internal recovery edges, and does not claim to separate almost all paired ports into different components or to eliminate longer paired paths.

The row-word proof depends only on the full head positions, raw-read positions, and append times, not on the semantic meaning of the shear's updates. Thus its relation-level code bound also applies if one changes the bounded-alphabet local rules while keeping exactly that trace. A cheap-genie fibre count would exclude a transposer on this specified trace: choose price less than ρL, fix the resulting depth, apply the full-code bound to each genie fibre, and sum the fibre cardinalities. We do not present that consequence as a new algorithm exclusion. The first paper's pass theorem [5, Theorem P] already excludes this fixed trace after absorbing its sublinear nonbank workspace into finite control at each dimension. The new content here is the uniform-over-genies, every-induced-code statement about the actual cone relation.

The fixed number of phases, full rewinds, monotone buffer-length sequences, small controller span, and common scan-event chart are essential verified properties of this example. No normalization of an arbitrary native schedule to these properties has been proved. In particular the theorem does not supply a deficit for all cheap constant-depth native cones, an adaptive common relation, or the general transposition lower bound.

7 What a transposition construction would need

The native examples distinguish inexpensive execution from inexpensive transposition. Expansion of the incidence graph and connected demand pairs can occur in linear time, but neither feature alone defeats a storage-code bound. Conversely, a construction cannot be excluded by a complex-linear theorem unless its actual genie fibres lie in the theorem's linear class with the stated coefficient and block-width bounds.

A candidate oblivious linear-time transposer must therefore fit both sides of the interface: its trace must evade the available cheap depth-reduction theorems, or its full finite-alphabet recovery relation must support enough codewords to absorb the cost of every applicable genie. The connected-demand example shows why a positive coding deficit uniform in fixed depth is useful for this comparison. It does not reduce an arbitrary candidate to the row interfaces analyzed here.

For an adaptive construction, fixing one trace is insufficient: any comparison must also account for the information needed to specify that trace and its recovery sets. The following section states these remaining obligations precisely.

8 The remaining coding problem

The structural theorem suggests a finite-alphabet question that mentions neither a transition function nor a choice of local encoding gates.

Question 8.1 (All-code paired-distance bound)**.** *Fix a degree bound Δ. Let N_j be recovery relations on finite coordinate sets V_j with $|V_j| \to \infty$. Suppose their incidence graphs have maximum degree at most Δ and satisfy the paired-distance condition. Must every sequence of nonempty finite-alphabet storage codes $\mathcal{C}_j \subseteq \mathcal{A}_j^{V_j}$ satisfy*

$$\limsup_{j\to\infty} \frac{\log_{|\mathcal{A}_j|} |\mathcal{C}_j|}{|V_j|} \le \frac{1}{2}?$$

Here $|\mathcal{A}_j| \ge 2$; the alphabets may vary. The fixed binary alphabet is already an important open case.

The directed analytic theorem answers the linear, bounded-complex-coefficient version at fixed vector width. The hyperfinite theorem answers the all-code question under an additional separator hypothesis. The native examples show that this separator hypothesis is not a consequence of a

linear-time origin. Their separate coding bounds show, at the same time, that its failure is not evidence for a high-rate code.

Proposition 8.2 (What an affirmative answer would supply)**.** Not formalized; a complete proof is given below.

If Question 8.1 has an affirmative answer for every fixed Δ, then for every fixed tape count, linear-time constant, and cone depth, all finite-alphabet storage codes on the corresponding actual transposition-labelled cone relations have rate at most $1/2 + o(1)$, uniformly over those schedules and genies.

Proof (not formalized). Let B be the fixed upper bound on recovery-set sizes. For any fixed $0 < \delta < 1$, condition on the coordinates whose left incidence degree exceeds an integer $M \geq B/\delta$. There are at most δL such coordinates. The retained relation has maximum degree at most $\max(B, M)$, and the native structural theorem gives $b_E(r) = O_r(L/\log d) = o(L)$ for every fixed r. Its number of coordinates is at least $(1-\delta)L$, so the same condition holds relative to its size.

An affirmative answer gives half-rate plus $o(1)$ on the retained relation. By Lemma 2.2, a full code has logarithmic size at most

$$|H| + \tfrac{1}{2}(L - |H|) + o(L) \leq \tfrac{1}{2}L + \tfrac{1}{2}\delta L + o(L).$$

First take $d \to \infty$ with δ fixed, then let $\delta \downarrow 0$. Uniformity follows by contradiction: a violating sequence of schedules, genies, and codes would itself be a sequence to which Question 8.1 applies after the same fixed-threshold trimming. □

This implication requires no assertion that the input and output ports are disconnected in the finite graphs. It also does not require that the recovery relation be symmetric. Conversely, a counterexample to Question 8.1 would defeat this proposed abstract coding theorem, not construct a transposer: the counterexample might have no native realization.

8.1 Depth, rate, and the order of choices

The transposition application still needs a cheap genie. The fourth paper supplies it for home-based head schedules and certain other restricted motion classes, but the general depth-reduction problem remains open [4]. If the all-code half bound held, any such class admitting genies of an arbitrarily small fixed positive cost would be excluded from oblivious linear-time transposition.

The exact value $1/2$ is not essential to that comparison. A deficit $\rho > 0$ works if one can choose a genie of cost εL with $\varepsilon < \rho$ and then apply the coding bound at its resulting fixed depth. A bound $\rho(D) > 0$ for each D is not by itself enough: making the genie cheaper can increase D, and the deficit may then fall below its cost. This is why the depth-independent leading constant in the connected-demand example is useful, even though it is smaller than one half.

For the specific native examples, the conclusion concerns all codes on their full cone relations. Their schedules already lie within classes excluded by the first paper's pass bounds. Extending the relation-level theorem to more general schedules would be new work; the examples are not a reduction of arbitrary machines to a finite set of scans.

8.2 Finite fields and nonlinear recovery

The complex theorem controls neither modular rank collapse nor nonlinear recovery. Bounded integer lifts expose modular rank loss as torsion; they do not rule it out. The appendix's matrix certificates give an algebraic way to prove finite-field bounds without controlling coefficients or

vector width, but only when the required matrix labels exist. Establishing those labels for arbitrary paired-distance systems, or finding a different all-code inequality, would close a genuine remaining interface.

The linear-algebraic and nonlinear statements should therefore be kept separate. A machine's genie is an arbitrary function of its binary input even when its head schedule is fixed. No replacement of that function by a complex linear map is available from the geometry alone.

8.3 Adaptive schedules

The geometric incidence theorem applies to each fixed legal adaptive trace. Its proof transports abstract labels along frozen head walks and does not randomize the inputs that select the trace. To use it for an adaptive machine, however, one must control the family of inputs producing that trace and the information needed to identify the relevant recovery relation. Different inputs can have different cones, output times, and genies. The tracewise theorem supplies none of that description for free.

Thus the paper isolates a specific structural invariant and several coding mechanisms that respect it. A complete lower-bound argument still needs a sufficiently strong finite-alphabet capacity bound, a compatible cheap depth reduction, and, for unrestricted (T), a paid all-input treatment of adaptivity.

A Finite matrix certificates and an expanding example

The component phases in Theorem 4.5 have a finite algebraic counterpart. Its witnesses may have growing dimension while the code being bounded retains any chosen block width. This distinction permits a finite-field result without a spectral limit of recovery matrices.

Lemma A.1 (Rank identity)**.** `Lean: rank_identity` [`RankCertificate.lean`]

For endomorphisms A, L, R of any finite-dimensional vector space over any field F,

$$2\,\mathrm{rank}(I - A) \geq \mathrm{rank}(L - R) - \mathrm{rank}(AL - RA). \tag{A.1}$$

No invertibility or commutativity is required.

Outline of the formal proof. For $M = I - A$ the exact identity is $L - R = (AL - RA) + ML - RM$. Apply rank subadditivity and $\mathrm{rank}(ML), \mathrm{rank}(RM) \leq \mathrm{rank}\, M$. □

Proposition A.2 (Matrix-label certificate)**.** `Lean: matrix_label_certificate, matrix_label_certificate_extension` [`MatrixCertificate.lean`]

Let A have $t \times t$ blocks indexed by N coordinates, supported on a paired incidence relation E. Give its ports arbitrary labels $X_p, Y_q \in \mathrm{Mat}_m(F)$, where $m \geq 1$. Then

$$\mathrm{rank}_F(I - A) \geq \frac{t \sum_p \mathrm{rank}_F(X_p - Y_p) - \sum_{L_p R_q \in E} \mathrm{rank}_F(A_{qp})\, \mathrm{rank}_F(X_p - Y_q)}{2m}. \tag{A.2}$$

Labels over a field extension of F are also valid witnesses, with the ranks of the original matrices unchanged.

Outline of the formal proof. Use $\widehat{A} = A \otimes I_m$ and block-diagonal operators $L_{pp} = I_t \otimes X_p$, $R_{qq} = I_t \otimes Y_q$. Then

$$\mathrm{rank}(I - \widehat{A}) = m\,\mathrm{rank}(I - A), \qquad \mathrm{rank}(L - R) = t \sum_p \mathrm{rank}(X_p - Y_p),$$

and the (q,p) block of $\widehat{A}L - R\widehat{A}$ is $A_{qp} \otimes (X_p - Y_q)$. Tensor-product ranks multiply over any field. Writing a block matrix as the sum of its single-block inclusions bounds its rank by the sum of their ranks. Lemma A.1 gives the displayed inequality. Scalar extension preserves matrix rank, so the same argument applies to labels over any extension field. □

In particular, full-rank differences on matched ports and low-rank differences on incidence edges give a useful certificate. The following construction shows that this situation is compatible with uniform expansion and large matched distances.

Theorem A.3 (An existential characteristic-two family)**.** Not formalized; a complete proof is given below.

There are constants $\Delta \geq 1$, $h > 0$, and m_0 such that for every integer $m \geq m_0$ there is a simple paired incidence graph E_m on $2N_m$ ports, where

$$N_m = 2^{m^2}(2^m - 1)^2,$$

with the following properties:

1. *Its maximum degree is at most Δ, and every port set S with $|S| \leq N_m$ has at least $h|S|$ edges leaving it.*
2. *Every matched pair has distance at least m.*
3. *For every field F of characteristic two, every integer $t \geq 1$, and every recovery matrix A of block width t supported on E_m,*

$$\operatorname{rank}_F(I - A) \geq \frac{tN_m}{2} - \frac{1}{2m} \sum_{L_p R_q \in E_m} \operatorname{rank}_F(A_{qp}) \geq \left(\frac{1}{2} - \frac{\Delta}{2m}\right) tN_m. \tag{A.3}$$

The graph is chosen independently of F, t, A. In particular the bound is uniform in t, even if t grows with m.

Proof (not formalized). An outer graph with rank-one steps. Put $s = 2^m$ and $D_m = (s-1)^2$. On the additive group $\mathrm{Mat}_m(\mathbb{F}_2)$ join B to $B + uv^{\mathsf{T}}$ for each ordered pair of nonzero vectors $u, v \in \mathbb{F}_2^m$. Each nonzero rank-one matrix has exactly one such representation over $\mathbb{F}_2$, so this is a simple D_m-regular graph G_m. The edge label (u, v) is unchanged by reversal.

For a matrix C of rank r, the additive character $B \mapsto (-1)^{\sum_{i,j} C_{ij} B_{ij}}$ has normalized eigenvalue

$$\lambda_r = \frac{2^{2m-r} - 2^{m+1} + 1}{(2^m - 1)^2}. \tag{A.4}$$

Indeed, among the $s-1$ nonzero vectors u, exactly $2^{m-r}-1$ satisfy $u^{\mathsf{T}}C = 0$. Summing the character over nonzero v gives $s-1$ for those u and -1 for the others. The resulting sum is $2^{2m-r} - 2s + 1$. These characters form an orthogonal eigenbasis. For $m \geq 2$ and $r \geq 1$, the eigenvalues decrease from at most $1/2$ to $-1/(s-1) \geq -1/3$. Thus all nonconstant eigenvalues have absolute value at most $1/2$.

Reducing the degree. Fix $k = 1024$. Theorem 1 of Bordenave [3] implies that, for every sufficiently large D, there is a simple k-regular graph H_D on D vertices whose normalized nonconstant eigenvalues have absolute value at most $1/10$. In fact $(2\sqrt{1023}+1)/1024 < 1/10$, and kD is always even. Choose one such graph on the label set of G_m.

Use the zig-zag construction of Reingold, Vadhan, and Wigderson [9, Section 3]. Its vertex set consists of pairs (B, a), where a is an outer edge label. A step first changes a along one edge of H_{D_m}, then takes that labelled edge of G_m, and finally changes the new label along one edge of H_{D_m}. The resulting undirected multigraph Z_m has N_m vertices and degree $\Delta = k^2$. Every edge changes B by a matrix of rank exactly one. There are no loops; we permit parallel edges in the spectral accounting.

For completeness, the following loose spectral estimate suffices. Let P average within each label cloud, let $\mathcal{H}$ apply the normalized adjacency of H_{D_m} within clouds, and let $\mathcal{R}$ be the outer rotation, an orthogonal involution. The normalized adjacency of Z_m is $\mathcal{HRH}$. For a vector f orthogonal to the constants, write $f = f_\parallel + f_\perp$, where $f_\parallel = Pf$, and put $e = \mathcal{H}f_\perp$. Then $\|e\| \le \|f_\perp\|/10$. On cloud-constant vectors $P\mathcal{R}P$ is the outer normalized adjacency. Hence

$$|\langle f, \mathcal{HRH}f\rangle| \le \tfrac{1}{2}\|f_\parallel\|^2 + \tfrac{2}{10}\|f_\parallel\|\|f_\perp\| + \tfrac{1}{100}\|f_\perp\|^2 \le 0.71\|f\|^2.$$

Since the adjacency is self-adjoint, all its nonconstant eigenvalues have absolute value at most $\lambda = 0.71 < 1$.

Ports and their prescribed pairing. For a vertex $x = (B, a)$ put $\theta(x) = (B + I_m, a)$. Take the bipartite double cover of Z_m, keep the physical left port at x named L_x, and name the physical right port at $\theta(y)$ by R_y. Finally remove parallel copies of edges to obtain the simple graph E_m. Thus

$$L_x R_y \in E_m \quad \Longleftrightarrow \quad x \text{ and } \theta(y) \text{ are adjacent in } Z_m. \tag{A.5}$$

Give these ports the matrix labels $X_x = B_x$ and $Y_y = B_y + I_m$. Each edge difference has rank one. Each matched difference is I_m and has rank m. Along any port path the matrix differences telescope; rank subadditivity therefore forces every matched path to have at least m edges. In particular there are no self-demand edges when $m > 1$.

The double cover has eigenvalues equal to the positive and negative eigenvalues of Z_m. Its eigenvalue 1 is simple, and every other eigenvalue is at most λ; the additional eigenvalue -1 is harmless for this upper bound. For a port set S of size $q \le N_m$, apply the spectral gap to $1_S - q/(2N_m)$. It gives at least $\Delta(1-\lambda)q/2$ crossing edges counted with multiplicity. Every simple edge has multiplicity at most Δ. Deduplication therefore leaves at least hq crossing edges for $h = (1-\lambda)/2 > 0$, and degree at most Δ.

The rank bound. Interpret the binary labels X_x, Y_y in any field of characteristic two. Their ranks are preserved by extension of the prime field. Proposition A.2 gives the first inequality in (A.3). The second follows from $\operatorname{rank} A_{yx} \le t$ and $|E_m| \le \Delta N_m$. □

Corollary A.4 (Linear codes on the expanding family)**.** Lean (partial): `linear_recovery_matrix_exists`, `linear_code_rate_of_matrix_labels` [`LinearRecovery.lean`]
Not formalized: Construction of the expanding family and its full-rank matched and rank-one edge labels.

Over any characteristic-two field, every linear storage space on E_m with t symbols per coordinate has normalized dimension at most $1/2 + \Delta/(2m)$. This includes linear spaces initially supplied with nonlinear recovery functions.

Proof (not formalized). On a linear space, determination of a coordinate by its neighbours means that the kernel of the neighbour projection is contained in the kernel of the coordinate projection. The latter therefore factors linearly through the image of the former, and this linear map extends to the ambient neighbour space. Choose these linear recovery maps for all coordinates to obtain A with $\mathcal{C} \subseteq \ker(I - A)$, and use Theorem A.3. □

These graphs are not uniformly hyperfinite. If deletion of z ports leaves components of size at most a fixed $K < N_m$, expansion summed over the components gives $h(2N_m - z) \leq \Delta z$. Thus every such separator has a fixed positive linear size. The preceding finite-field result consequently goes beyond the hyperfinite hypothesis of Corollary 4.4, though it applies only to linear code spaces.

Two scope points are useful. First, the binary translation θ is an involutive automorphism of Z_m, so the demand relation in this particular construction is symmetric. Its contribution is a characteristic-two bound uniform in block width, not an example requiring asymmetric recovery. Second, the inner graphs were chosen by an existence theorem. No efficient uniform generator or native tape realization is asserted. These are abstract relations; their degree, expansion, and paired-port separation do not identify them with all native cones.

B Index of Lean declarations

Declarations are relative to `DirectedRecovery`; their axiom checks are generated together with this index (Appendix C).

DirectedRecovery.ColorOrder `color_order_half_bound`.

DirectedRecovery.DamagedExpanders `DamagedExpanders.damaged_expander_copies`.

DirectedRecovery.Hyperfinite `UniformlyHyperfinite`, `hyperfinite_rate_limsup_le_half`.

DirectedRecovery.IncidenceComponents `storageCode_card_le_of_separated_components`.

DirectedRecovery.IncidenceSeparator `storageCode_separator_log_bound`.

DirectedRecovery.IncidenceTrimming `paid_degree_trimming`.

DirectedRecovery.IndependentRectangle `IndependentRectangle.independent_rectangle`, `IndependentRectangle.independent_rectangle_consecutive`.

DirectedRecovery.LinearRecovery `linear_code_rate_of_matrix_labels`, `linear_recovery_matrix_exists`.

DirectedRecovery.MatrixCertificate `matrix_label_certificate`, `matrix_label_certificate_extension`.

DirectedRecovery.PathPackingAsymptotics `short_paired_count_log_asymptotics`, `short_paired_count_log_bound`.

DirectedRecovery.RankCertificate `rank_identity`.

DirectedRecovery.RowWordDeficit `row_word_cover_asymptotic_code_gap`, `row_word_cover_code_log_bound`, `row_word_cover_independent`.

DirectedRecovery.SelfDemands `self_demands_count_of_disjoint_transport`.

DirectedRecovery.StorageCode indexCode_length_ge, induced_storageCode_card_le_of_independent, induced_storageCode_entropy_of_independent, isStorageCode_conditionedCode, isStorageCode_indexFiber, storageCode_conditioning_bound.

DirectedRecovery.ThresholdBuckets ThresholdPullback.finite_threshold_pullback.

C Proof status and verification

Every numbered result carries an explicit status line. The finite conditioning, fibre, independent-coordinate, rectangle, threshold-pullback, port-color, component, separator, damaged-expander, rank-identity, and matrix-label bounds, together with the hyperfinite half-rate corollary, are Lean-verified. Finite path packing and its logarithmic asymptotic consequence are formalized from the native disjoint-path transport estimates; their derivation from tape semantics and their uniform constants remain written proofs. The self-demand count is formalized from the native one-edge transport bound, whose proof remains unformalized. The finite counting part of degree trimming is formalized; its native premise and asymptotic application remain written proofs. The finite and asymptotic depth-independent code deficit is formalized from an actual row-word cover, including canonical bucket selection; the native extraction and its uniform sublinear exception bound remain written proofs. The linear-code corollary is formalized from actual matrix-label witnesses, including linearization of nonlinear recovery; constructing the expanding family and its witnesses remains a written proof. The remaining new results currently have written proofs. The earlier papers are cited at a pinned revision; their verification claims apply only to their own stated declarations. The geometric, analytic and native-example arguments here are written proofs, subject to further scrutiny. The general all-code paired-distance bound, unrestricted cheap depth reduction, and an all-input treatment of adaptive schedules remain open.

The Lean library uses Lean and Mathlib v4.34.1 and requires the shared `lean-machine` package and the fourth paper's library from the same repository. No new axioms, `sorry`, `admit`, or kernel-bypassing decision procedures are permitted. The only permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. The declaration index and `PaperAxioms.lean` are generated from the same inline formal-status annotations by `tools/lean_index.py`.

From the paper directory, run `python3 tools/lean_index.py` to regenerate these files. In `lean/`, run `lake exe cache get`, `lake build`, and `lake env lean PaperAxioms.lean`. The entire repository must be present with its sibling directories. In `paper/`, run `latexmk -pdf main.tex` to compile the manuscript.

Formal verification, where recorded, establishes the precise Lean statements; it does not certify external novelty or remove hypotheses from their applications.

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